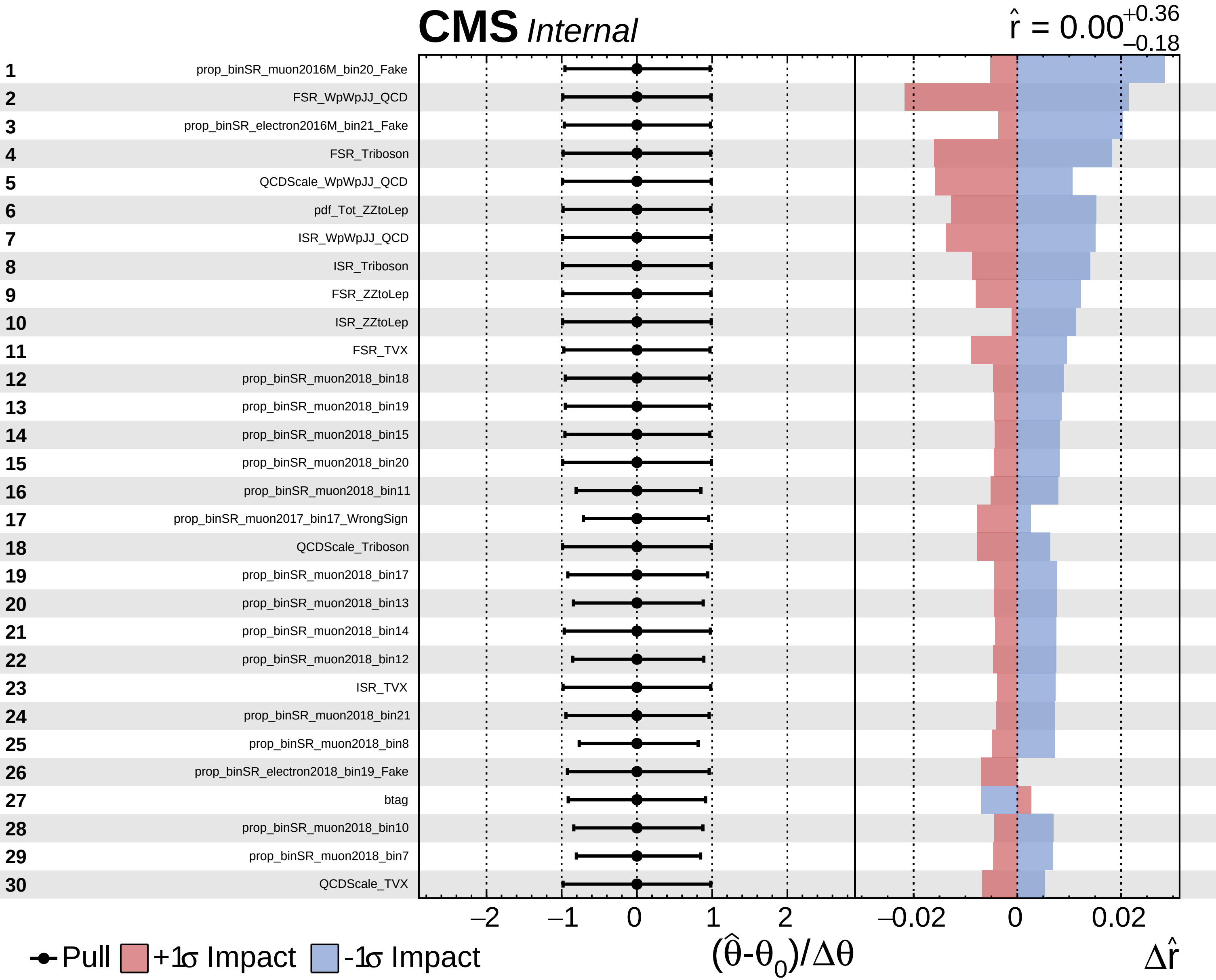
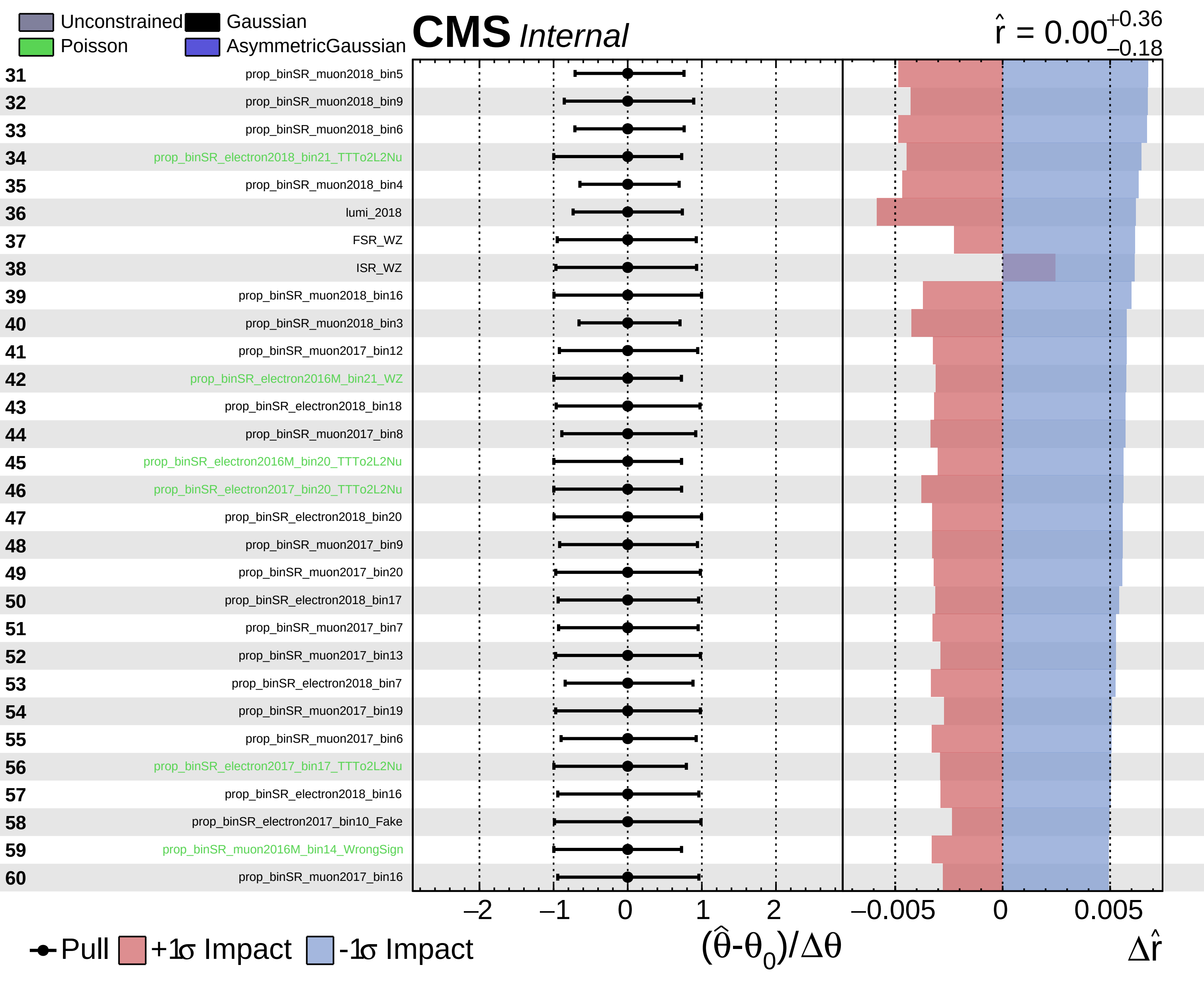
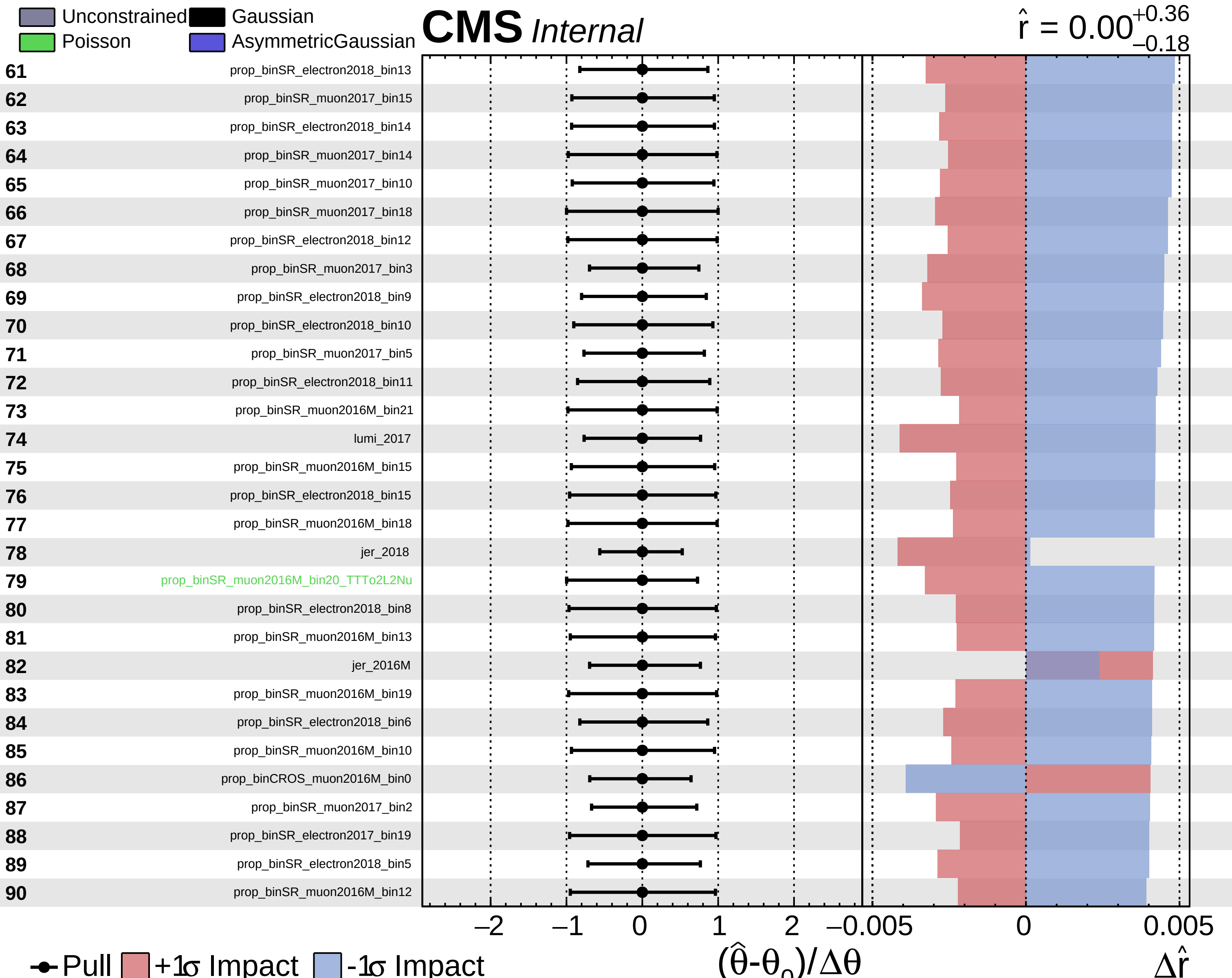


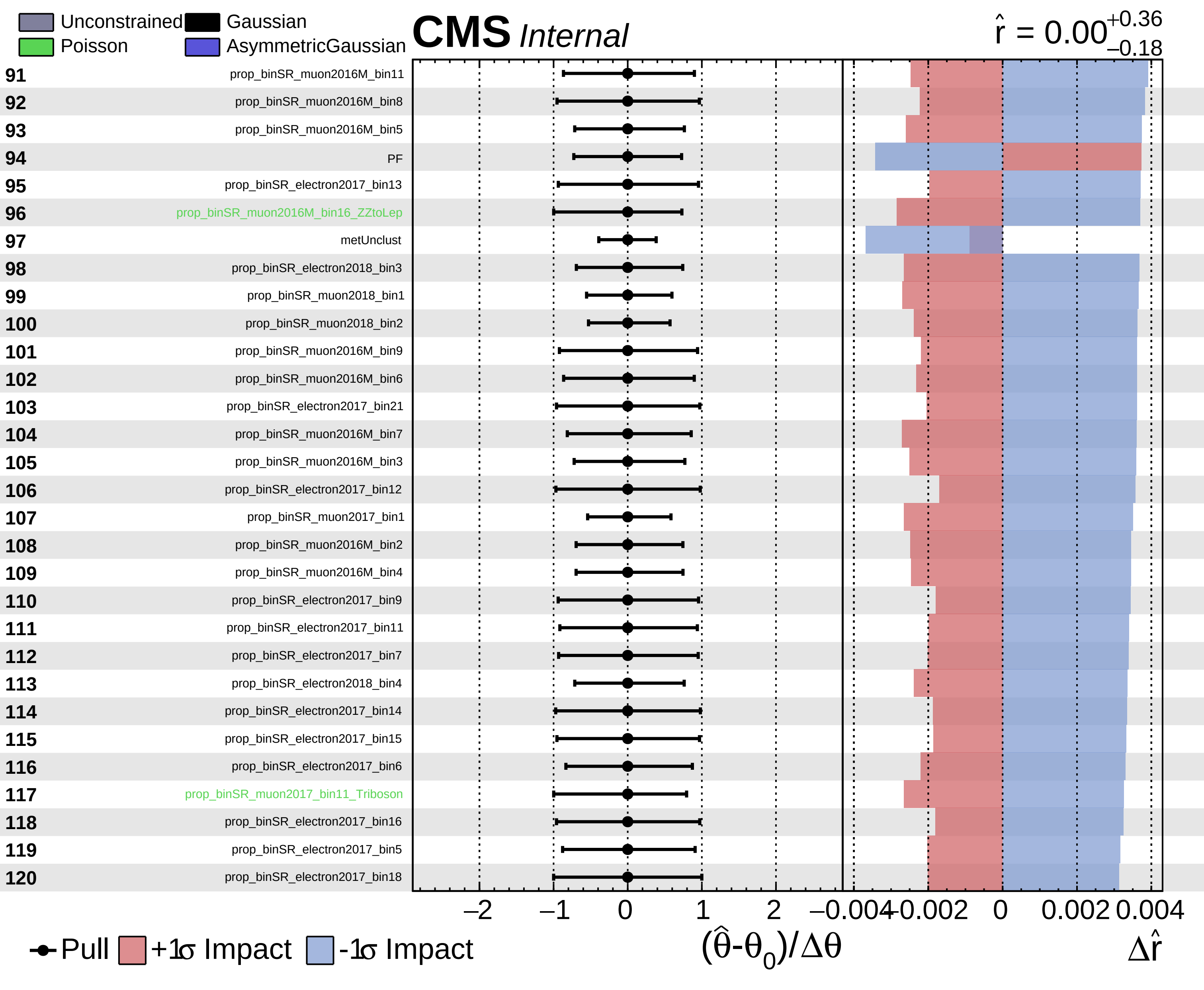
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.18}$
1 prop_binSR_muon2016M_bin20_Fake
2 FSR_WpWpJJ_QCD
3 prop_binSR_electron2016M_bin21_Fake
4 FSR_Triboson
5 QCDScale_WpWpJJ_QCD
6 pdf_Tot_ZZtoLep
7 ISR_WpWpJJ_QCD
8 ISR_Triboson
9 FSR_ZZtoLep
10 ISR_ZZtoLep
11 FSR_TVX
12 prop_binSR_muon2018_bin18
13 prop_binSR_muon2018_bin19
14 prop_binSR_muon2018_bin15
15 prop_binSR_muon2018_bin20
16 prop_binSR_muon2018_bin11
17 prop_binSR_muon2017_bin17_WrongSign
18 QCDScale_Triboson
19 prop_binSR_muon2018_bin17
20 prop_binSR_muon2018_bin13
21 prop_binSR_muon2018_bin14
22 prop_binSR_muon2018_bin12
23 ISR_TVX
24 prop_binSR_muon2018_bin21
25 prop_binSR_muon2018_bin8
26 prop_binSR_electron2018_bin19_Fake
27 btag
28 prop_binSR_muon2018_bin10
29 prop_binSR_muon2018_bin7
30 QCDScale_TVX
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



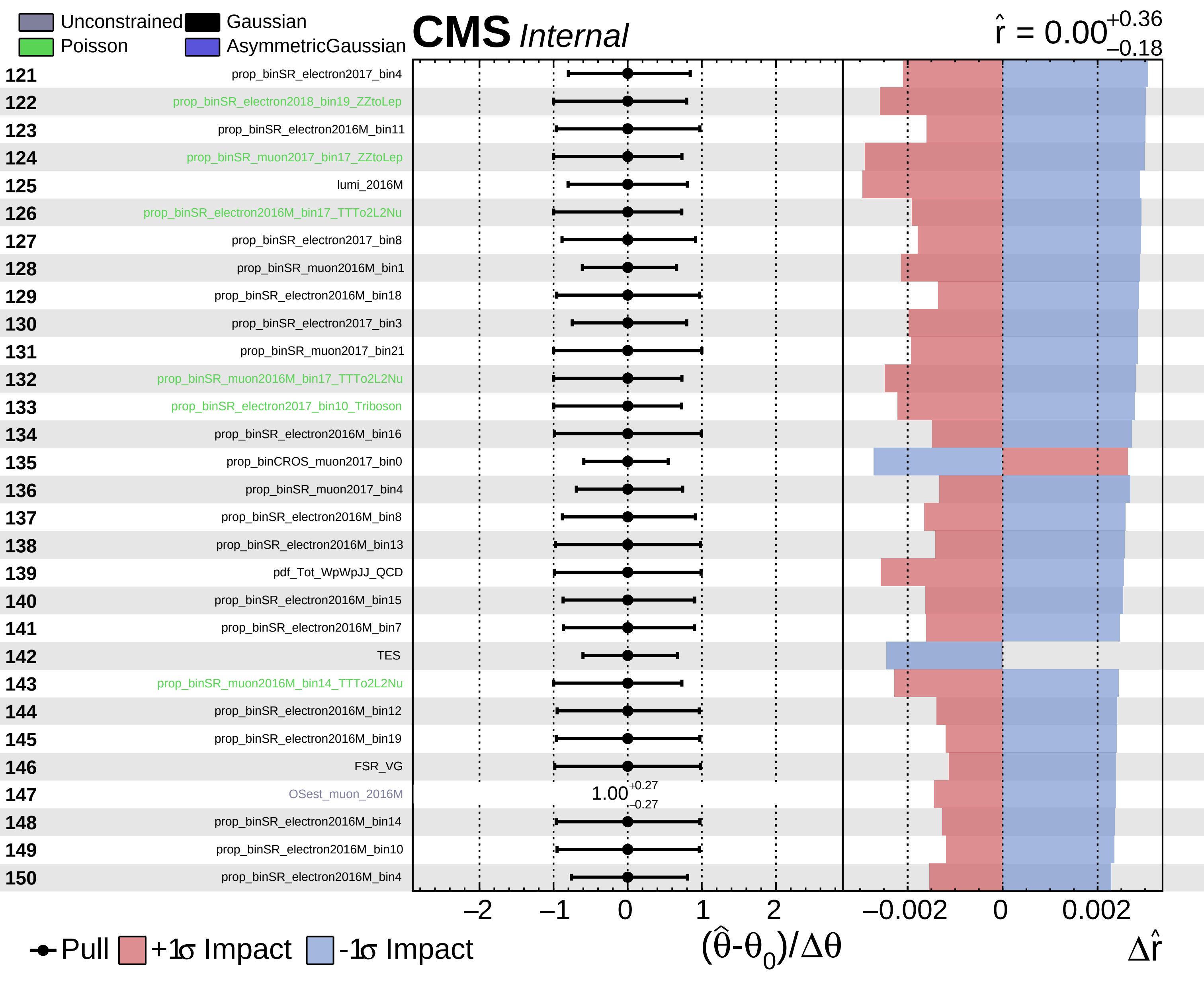
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.18}$
31 prop_binSR_muon2018_bin5
32 prop_binSR_muon2018_bin9
33 prop_binSR_muon2018_bin6
34 prop_binSR_electron2018_bin21_TTTo2L2Nu
35 prop_binSR_muon2018_bin4
36 lumi_2018
37 FSR_WZ
38 ISR_WZ
39 prop_binSR_muon2018_bin16
40 prop_binSR_muon2018_bin3
41 prop_binSR_muon2017_bin12
42 prop_binSR_electron2016M_bin21_WZ
43 prop_binSR_electron2018_bin18
44 prop_binSR_muon2017_bin8
45 prop_binSR_electron2016M_bin20_TTTo2L2Nu
46 prop_binSR_electron2017_bin20_TTTo2L2Nu
47 prop_binSR_electron2018_bin20
48 prop_binSR_muon2017_bin9
49 prop_binSR_muon2017_bin20
50 prop_binSR_electron2018_bin17
51 prop_binSR_muon2017_bin7
52 prop_binSR_muon2017_bin13
53 prop_binSR_electron2018_bin7
54 prop_binSR_muon2017_bin19
55 prop_binSR_muon2017_bin6
56 prop_binSR_electron2017_bin17_TTTo2L2Nu
57 prop_binSR_electron2018_bin16
58 prop_binSR_electron2017_bin10_Fake
59 prop_binSR_muon2016M_bin14_WrongSign
60 prop_binSR_muon2017_bin16
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

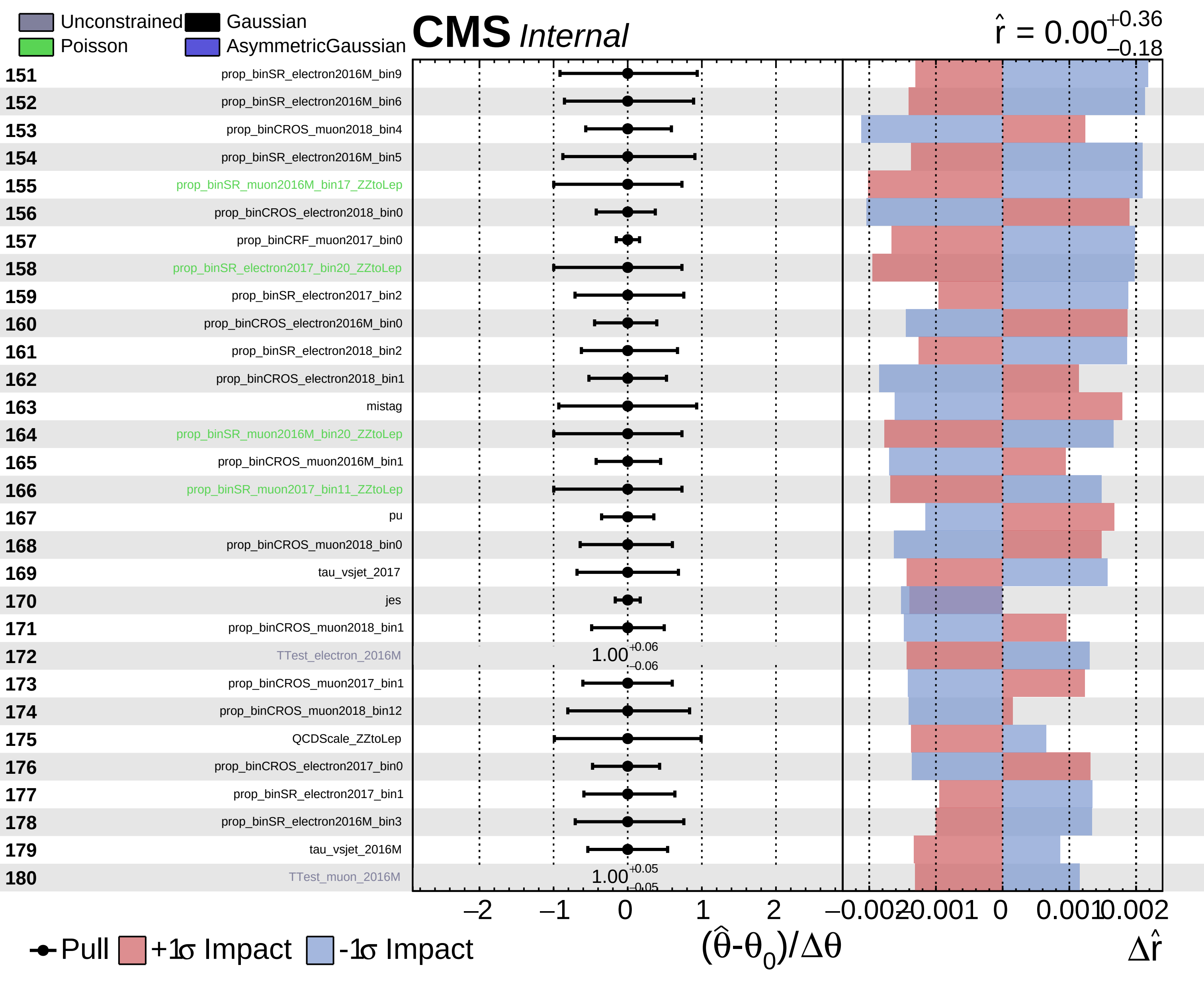


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.18}$
61 prop_binSR_electron2018_bin13
62 prop_binSR_muon2017_bin15
63 prop_binSR_electron2018_bin14
64 prop_binSR_muon2017_bin14
65 prop_binSR_muon2017_bin10
66 prop_binSR_muon2017_bin18
67 prop_binSR_electron2018_bin12
68 prop_binSR_muon2017_bin3
69 prop_binSR_electron2018_bin9
70 prop_binSR_electron2018_bin10
71 prop_binSR_muon2017_bin5
72 prop_binSR_electron2018_bin11
73 prop_binSR_muon2016M_bin21
74 lumi_2017
75 prop_binSR_muon2016M_bin15
76 prop_binSR_electron2018_bin15
77 prop_binSR_muon2016M_bin18
78 jer_2018
79 prop_binSR_muon2016M_bin20_TTTo2L2Nu
80 prop_binSR_electron2018_bin8
81 prop_binSR_muon2016M_bin13
82 jer_2016M
83 prop_binSR_muon2016M_bin19
84 prop_binSR_electron2018_bin6
85 prop_binSR_muon2016M_bin10
86 prop_binCROS_muon2016M_bin0
87 prop_binSR_muon2017_bin2
88 prop_binSR_electron2017_bin19
89 prop_binSR_electron2018_bin5
90 prop_binSR_muon2016M_bin12
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

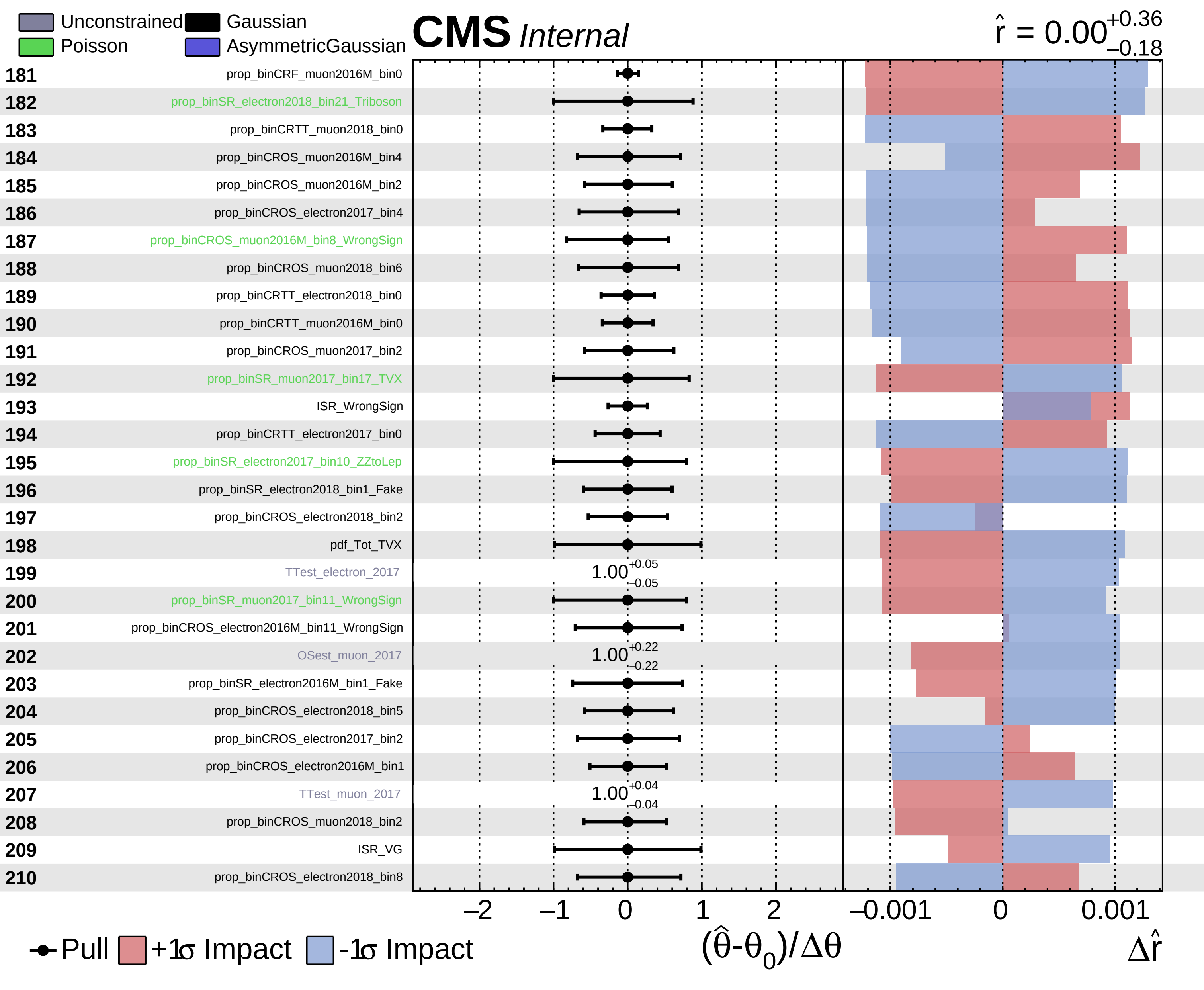


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.18}$
91 prop_binSR_muon2016M_bin11
92 prop_binSR_muon2016M_bin8
93 prop_binSR_muon2016M_bin5
94 PF
95 prop_binSR_electron2017_bin13
96 prop_binSR_muon2016M_bin16_ZZtoLep
97 metUnclust
98 prop_binSR_electron2018_bin3
99 prop_binSR_muon2018_bin1
100 prop_binSR_muon2018_bin2
101 prop_binSR_muon2016M_bin9
102 prop_binSR_muon2016M_bin6
103 prop_binSR_electron2017_bin21
104 prop_binSR_muon2016M_bin7
105 prop_binSR_muon2016M_bin3
106 prop_binSR_electron2017_bin12
107 prop_binSR_muon2017_bin1
108 prop_binSR_muon2016M_bin2
109 prop_binSR_muon2016M_bin4
110 prop_binSR_electron2017_bin9
111 prop_binSR_electron2017_bin11
112 prop_binSR_electron2017_bin7
113 prop_binSR_electron2018_bin4
114 prop_binSR_electron2017_bin14
115 prop_binSR_electron2017_bin15
116 prop_binSR_electron2017_bin6
117 prop_binSR_muon2017_bin11_Triboson
118 prop_binSR_electron2017_bin16
119 prop_binSR_electron2017_bin5
120 prop_binSR_electron2017_bin18
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

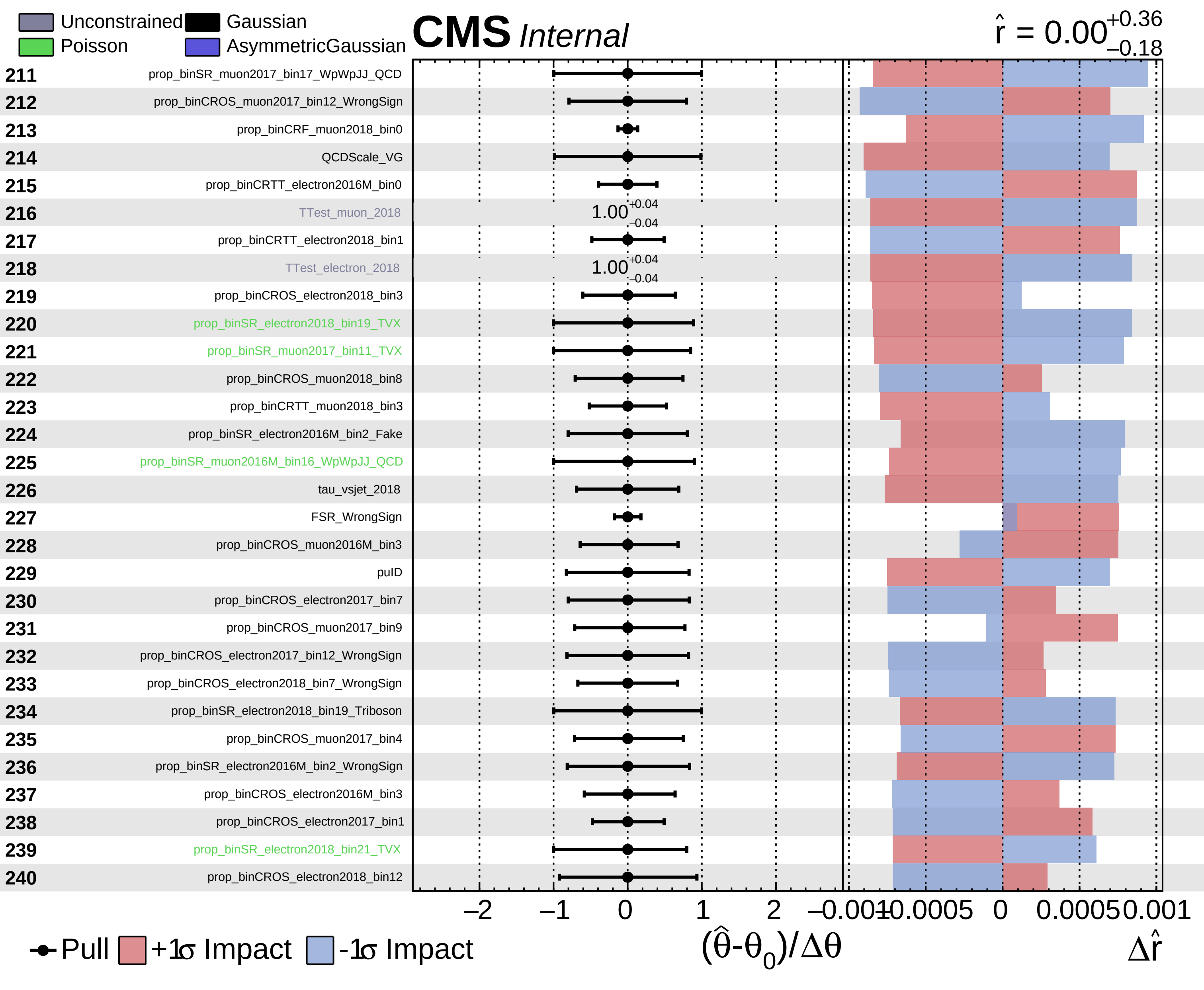

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.18}$
121 prop_binSR_electron2017_bin4
122 prop_binSR_electron2018_bin19_ZZtoLep
123 prop_binSR_electron2016M_bin11
124 prop_binSR_muon2017_bin17_ZZtoLep
125 lumi_2016M
126 prop_binSR_electron2016M_bin17_TTTo2L2Nu
127 prop_binSR_electron2017_bin8
128 prop_binSR_muon2016M_bin1
129 prop_binSR_electron2016M_bin18
130 prop_binSR_electron2017_bin3
131 prop_binSR_muon2017_bin21
132 prop_binSR_muon2016M_bin17_TTTo2L2Nu
133 prop_binSR_electron2017_bin10_Triboson
134 prop_binSR_electron2016M_bin16
135 prop_binCROS_muon2017_bin0
136 prop_binSR_muon2017_bin4
137 prop_binSR_electron2016M_bin8
138 prop_binSR_electron2016M_bin13
139 pdf_Tot_WpWpJJ_QCD
140 prop_binSR_electron2016M_bin15
141 prop_binSR_electron2016M_bin7
142 TES
143 prop_binSR_muon2016M_bin14_TTTo2L2Nu
144 prop_binSR_electron2016M_bin12
145 prop_binSR_electron2016M_bin19
146 FSR_VG
147 OSest_muon_2016M
$1.00^{+0.27}_{-0.27}$
148 prop_binSR_electron2016M_bin14
149 prop_binSR_electron2016M_bin10
150 prop_binSR_electron2016M_bin4
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



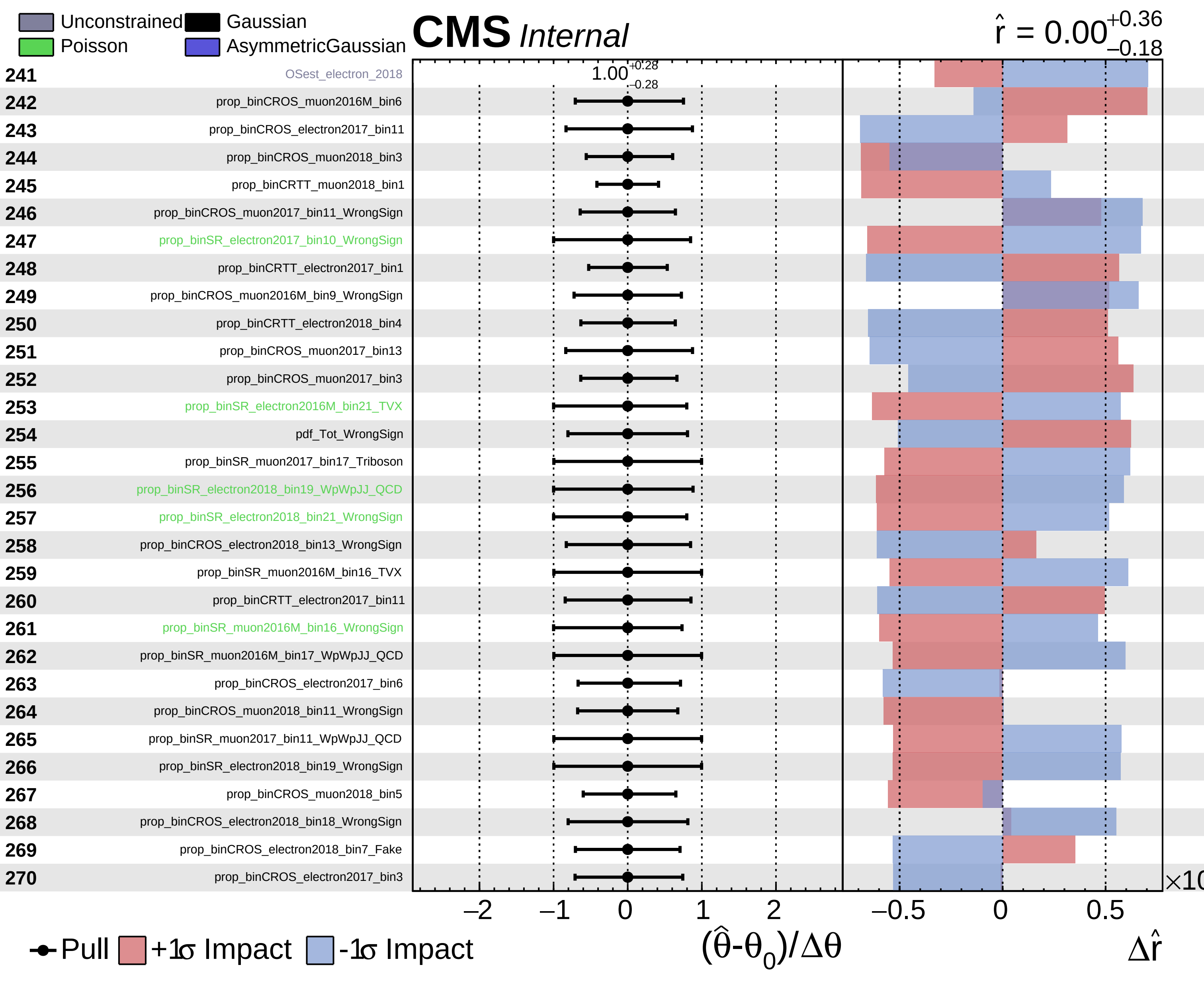
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.18}$
151 prop_binSR_electron2016M_bin9
152 prop_binSR_electron2016M_bin6
153 prop_binCROS_muon2018_bin4
154 prop_binSR_electron2016M_bin5
155 prop_binSR_muon2016M_bin17_ZZtoLep
156 prop_binCROS_electron2018_bin0
157 prop_binCRF_muon2017_bin0
158 prop_binSR_electron2017_bin20_ZZtoLep
159 prop_binSR_electron2017_bin2
160 prop_binCROS_electron2016M_bin0
161 prop_binSR_electron2018_bin2
162 prop_binCROS_electron2018_bin1
163 mistag
164 prop_binSR_muon2016M_bin20_ZZtoLep
165 prop_binCROS_muon2016M_bin1
166 prop_binSR_muon2017_bin11_ZZtoLep
167 pu
168 prop_binCROS_muon2018_bin0
169 tau_vsjet_2017
170 jes
171 prop_binCROS_muon2018_bin1
172 TTest_electron_2016M $1.00^{+0.06}_{-0.06}$
173 prop_binCROS_muon2017_bin1
174 prop_binCROS_muon2018_bin12
175 QCDScale_ZZtoLep
176 prop_binCROS_electron2017_bin0
177 prop_binSR_electron2017_bin1
178 prop_binSR_electron2016M_bin3
179 tau_vsjet_2016M
180 TTest_muon_2016M $1.00^{+0.05}_{-0.05}$
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



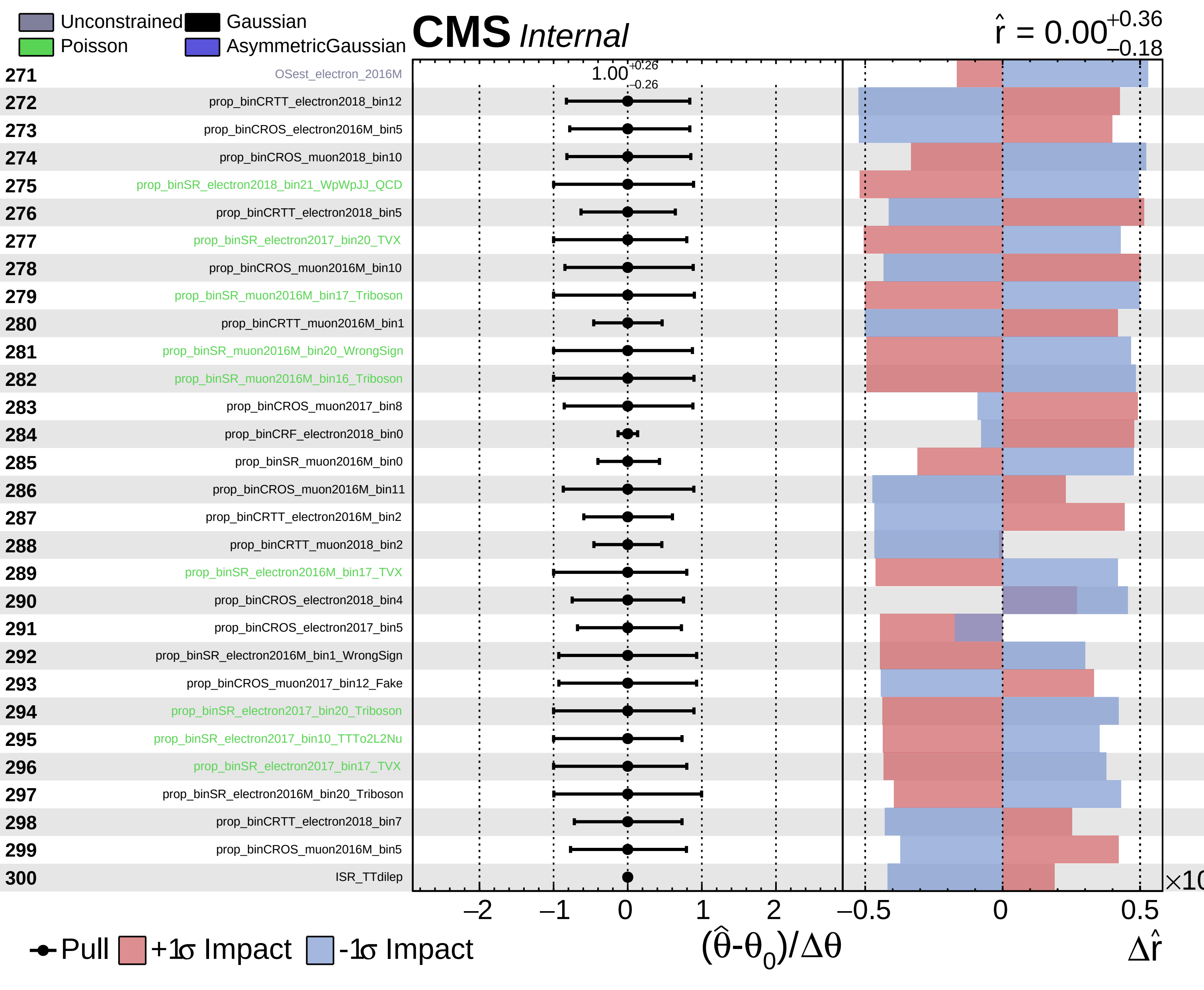
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.36 −0.18
181 prop_binCRF_muon2016M_bin0
182 prop_binSR_electron2018_bin21_Triboson
183 prop_binCRTT_muon2018_bin0
184 prop_binCROS_muon2016M_bin4
185 prop_binCROS_muon2016M_bin2
186 prop_binCROS_electron2017_bin4
187 prop_binCROS_muon2016M_bin8_WrongSign
188 prop_binCROS_muon2018_bin6
189 prop_binCRTT_electron2018_bin0
190 prop_binCRTT_muon2016M_bin0
191 prop_binCROS_muon2017_bin2
192 prop_binSR_muon2017_bin17_TVX
193 ISR_WrongSign
194 prop_binCRTT_electron2017_bin0
195 prop_binSR_electron2017_bin10_ZZtoLep
196 prop_binSR_electron2018_bin1_Fake
197 prop_binCROS_electron2018_bin2
198 pdf_Tot_TVX
199 TTest_electron_2017 1.00 +0.05 −0.05
200 prop_binSR_muon2017_bin11_WrongSign
201 prop_binCROS_electron2016M_bin11_WrongSign
202 OSest_muon_2017 1.00 +0.22 −0.22
203 prop_binSR_electron2016M_bin1_Fake
204 prop_binCROS_electron2018_bin5
205 prop_binCROS_electron2017_bin2
206 prop_binCROS_electron2016M_bin1
207 TTest_muon_2017 1.00 +0.04 −0.04
208 prop_binCROS_muon2018_bin2
209 ISR_VG
210 prop_binCROS_electron2018_bin8
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



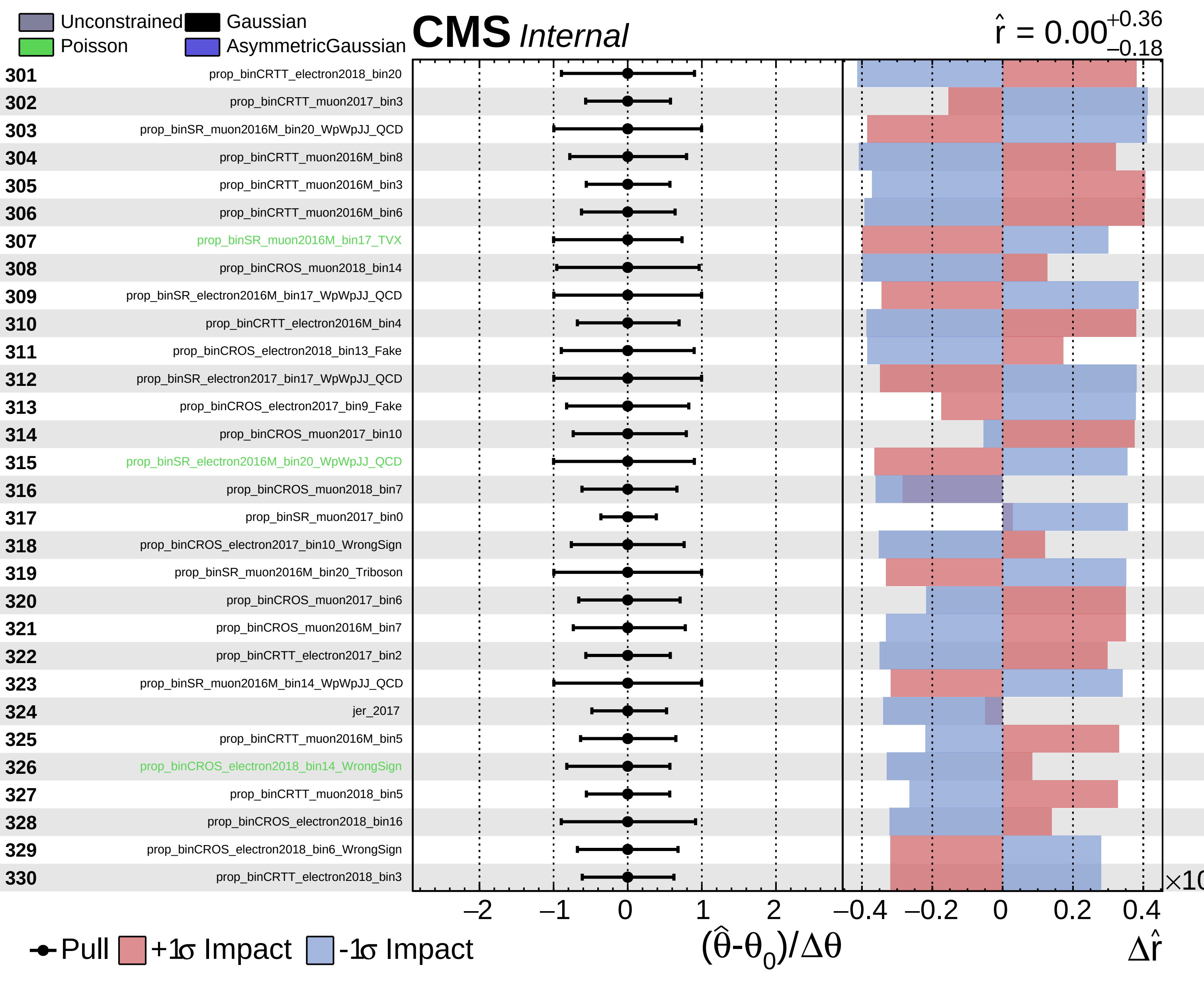
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.18}$
211 prop_binSR_muon2017_bin17_WpWpJJ_QCD
212 prop_binCROS_muon2017_bin12_WrongSign
213 prop_binCRF_muon2018_bin0
214 QCDScale_VG
215 prop_binCRTT_electron2016M_bin0
216 TTest_muon_2018
$1.00^{+0.04}_{-0.04}$
217 prop_binCRTT_electron2018_bin1
218 TTest_electron_2018
$1.00^{+0.04}_{-0.04}$
219 prop_binCROS_electron2018_bin3
220 prop_binSR_electron2018_bin19_TVX
221 prop_binSR_muon2017_bin11_TVX
222 prop_binCROS_muon2018_bin8
223 prop_binCRTT_muon2018_bin3
224 prop_binSR_electron2016M_bin2_Fake
225 prop_binSR_muon2016M_bin16_WpWpJJ_QCD
226 tau_vsjet_2018
227 FSR_WrongSign
228 prop_binCROS_muon2016M_bin3
229 puID
230 prop_binCROS_electron2017_bin7
231 prop_binCROS_muon2017_bin9
232 prop_binCROS_electron2017_bin12_WrongSign
233 prop_binCROS_electron2018_bin7_WrongSign
234 prop_binSR_electron2018_bin19_Triboson
235 prop_binCROS_muon2017_bin4
236 prop_binSR_electron2016M_bin2_WrongSign
237 prop_binCROS_electron2016M_bin3
238 prop_binCROS_electron2017_bin1
239 prop_binSR_electron2018_bin21_TVX
240 prop_binCROS_electron2018_bin12
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

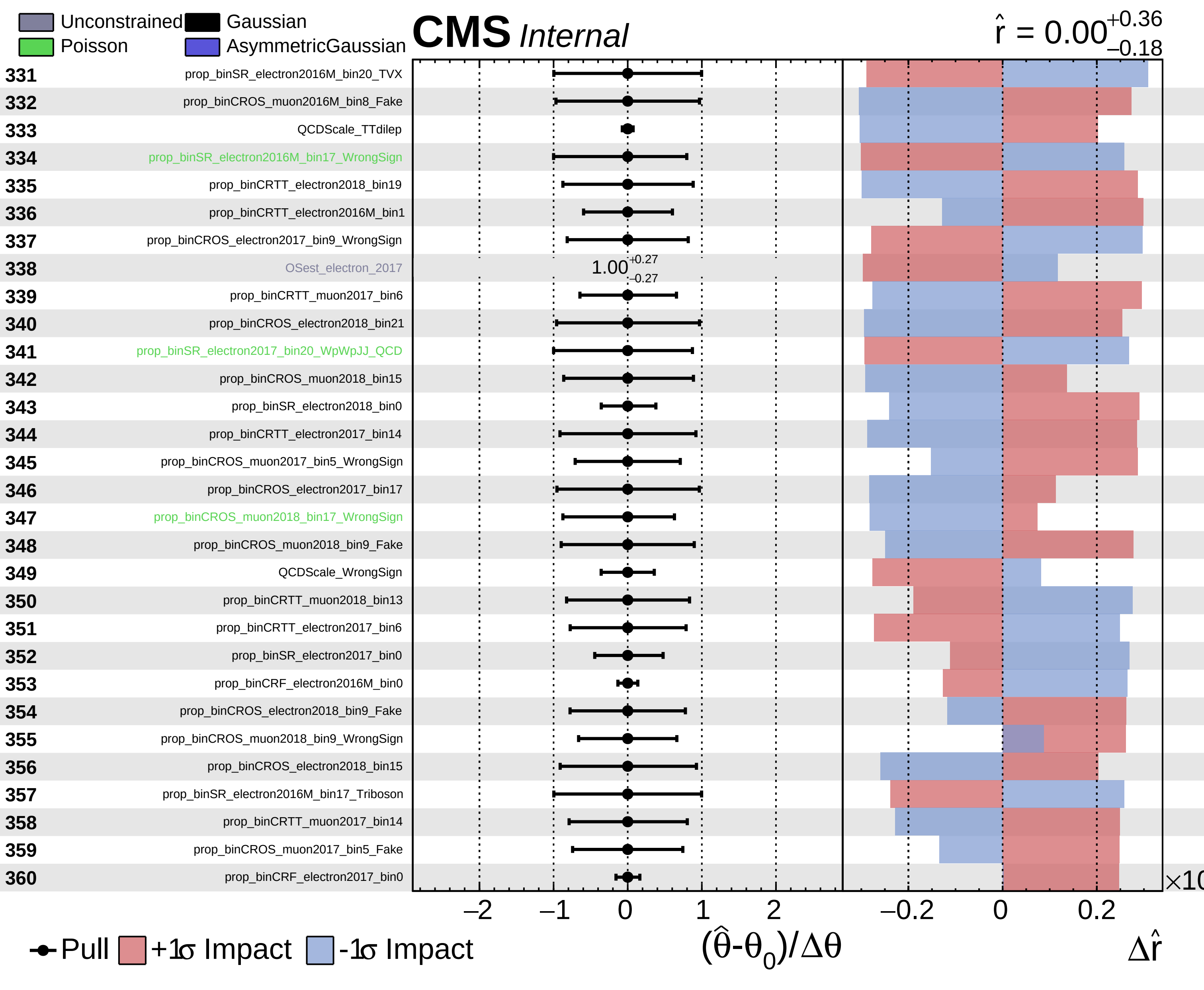


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.18}$
241 OSest_electron_2018
$1.00^{+0.28}_{-0.28}$
242 prop_binCROS_muon2016M_bin6
243 prop_binCROS_electron2017_bin11
244 prop_binCROS_muon2018_bin3
245 prop_binCRTT_muon2018_bin1
246 prop_binCROS_muon2017_bin11_WrongSign
247 prop_binSR_electron2017_bin10_WrongSign
248 prop_binCRTT_electron2017_bin1
249 prop_binCROS_muon2016M_bin9_WrongSign
250 prop_binCRTT_electron2018_bin4
251 prop_binCROS_muon2017_bin13
252 prop_binCROS_muon2017_bin3
253 prop_binSR_electron2016M_bin21_TVX
254 pdf_Tot_WrongSign
255 prop_binSR_muon2017_bin17_Triboson
256 prop_binSR_electron2018_bin19_WpWpJJ_QCD
257 prop_binSR_electron2018_bin21_WrongSign
258 prop_binCROS_electron2018_bin13_WrongSign
259 prop_binSR_muon2016M_bin16_TVX
260 prop_binCRTT_electron2017_bin11
261 prop_binSR_muon2016M_bin16_WrongSign
262 prop_binSR_muon2016M_bin17_WpWpJJ_QCD
263 prop_binCROS_electron2017_bin6
264 prop_binCROS_muon2018_bin11_WrongSign
265 prop_binSR_muon2017_bin11_WpWpJJ_QCD
266 prop_binSR_electron2018_bin19_WrongSign
267 prop_binCROS_muon2018_bin5
268 prop_binCROS_electron2018_bin18_WrongSign
269 prop_binCROS_electron2018_bin7_Fake
270 prop_binCROS_electron2017_bin3
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

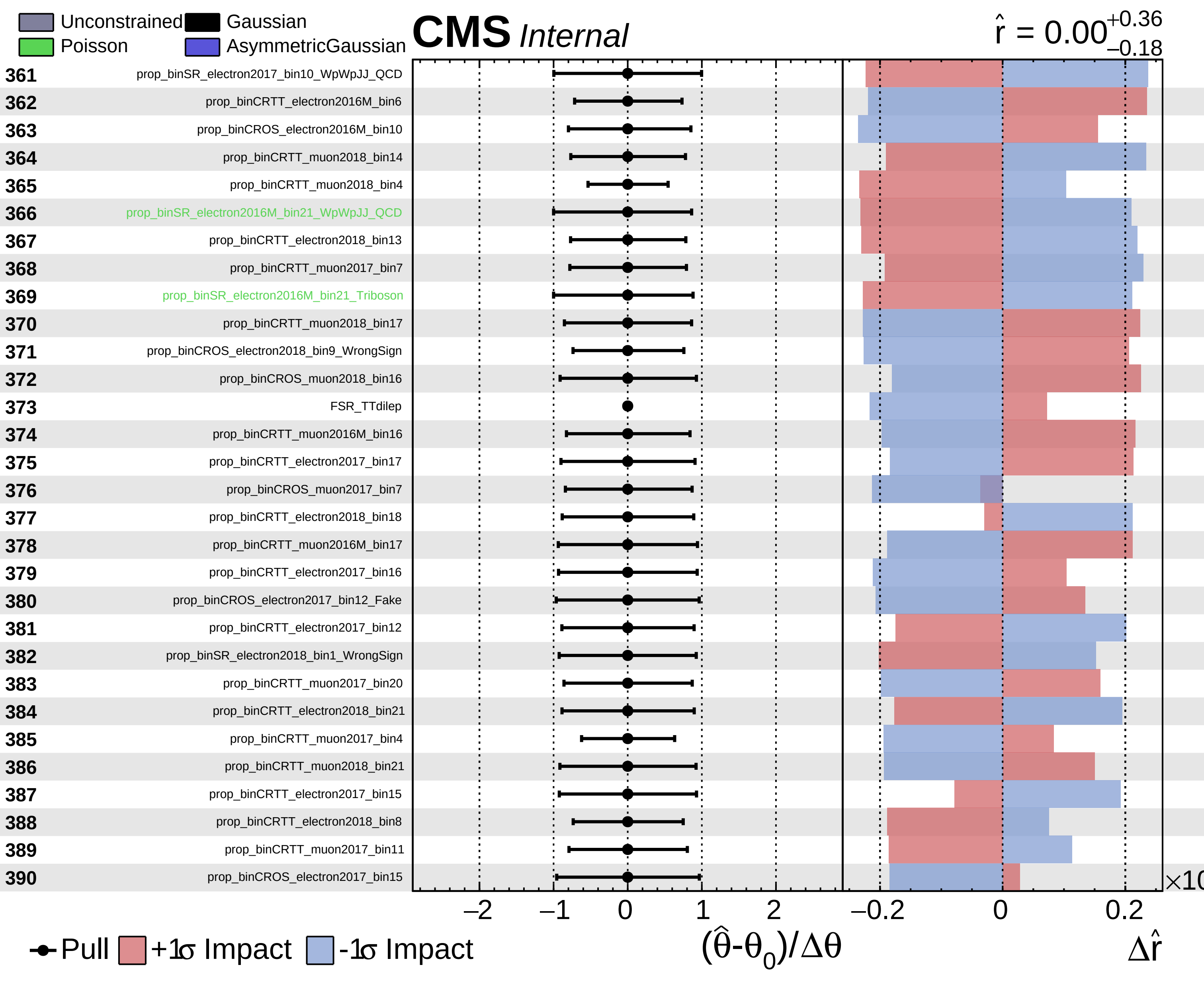


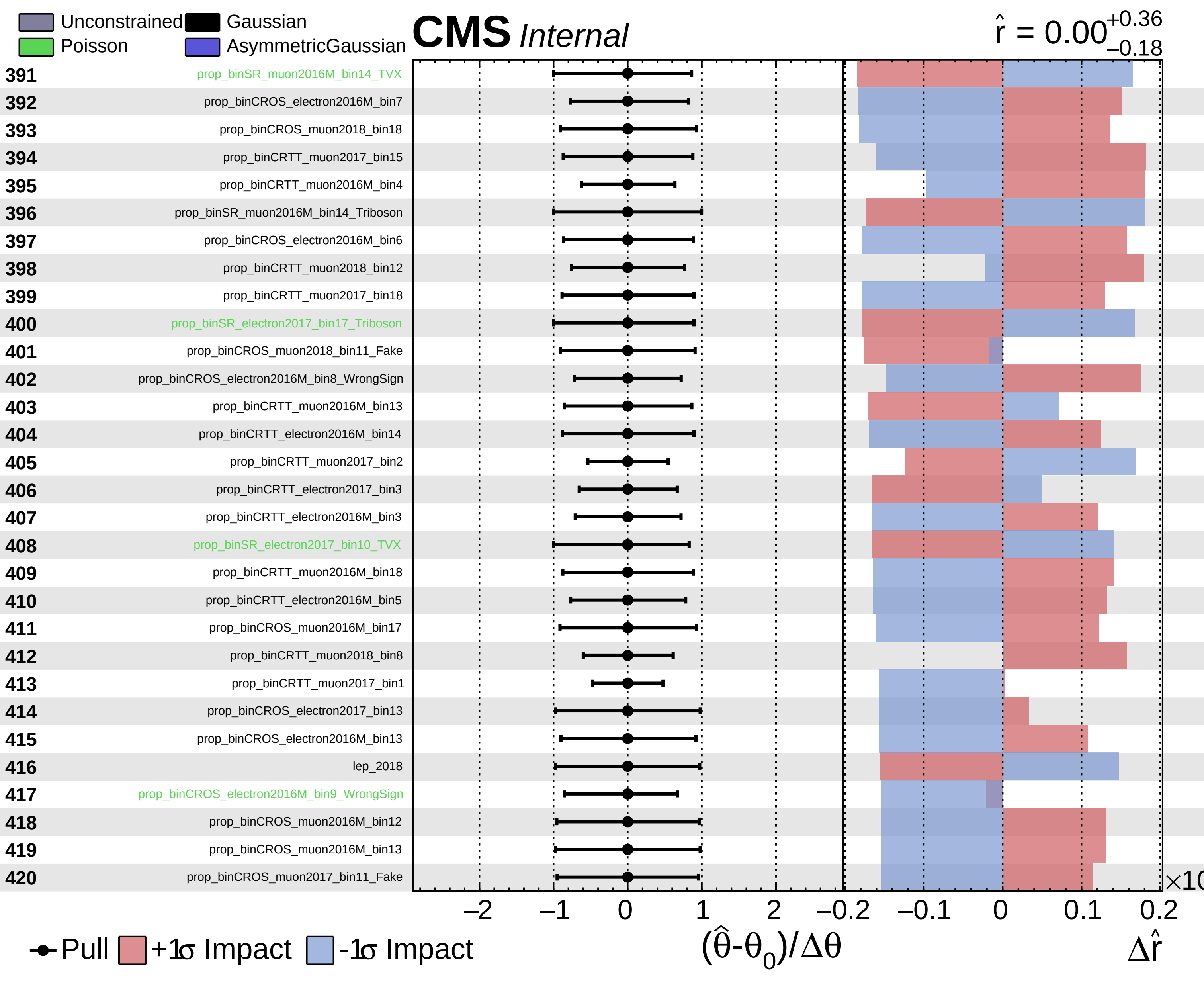
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.18}$
271 OSest_electron_2016M
$1.00^{+0.26}_{-0.26}$
272 prop_binCRTT_electron2018_bin12
273 prop_binCROS_electron2016M_bin5
274 prop_binCROS_muon2018_bin10
275 prop_binSR_electron2018_bin21_WpWpJJ_QCD
276 prop_binCRTT_electron2018_bin5
277 prop_binSR_electron2017_bin20_TVX
278 prop_binCROS_muon2016M_bin10
279 prop_binSR_muon2016M_bin17_Triboson
280 prop_binCRTT_muon2016M_bin1
281 prop_binSR_muon2016M_bin20_WrongSign
282 prop_binSR_muon2016M_bin16_Triboson
283 prop_binCROS_muon2017_bin8
284 prop_binCRF_electron2018_bin0
285 prop_binSR_muon2016M_bin0
286 prop_binCROS_muon2016M_bin11
287 prop_binCRTT_electron2016M_bin2
288 prop_binCRTT_muon2018_bin2
289 prop_binSR_electron2016M_bin17_TVX
290 prop_binCROS_electron2018_bin4
291 prop_binCROS_electron2017_bin5
292 prop_binSR_electron2016M_bin1_WrongSign
293 prop_binCROS_muon2017_bin12_Fake
294 prop_binSR_electron2017_bin20_Triboson
295 prop_binSR_electron2017_bin10_TTTo2L2Nu
296 prop_binSR_electron2017_bin17_TVX
297 prop_binSR_electron2016M_bin20_Triboson
298 prop_binCRTT_electron2018_bin7
299 prop_binCROS_muon2016M_bin5
300 ISR_TTdilep
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.36 −0.18
301 prop_binCRTT_electron2018_bin20
302 prop_binCRTT_muon2017_bin3
303 prop_binSR_muon2016M_bin20_WpWpJJ_QCD
304 prop_binCRTT_muon2016M_bin8
305 prop_binCRTT_muon2016M_bin3
306 prop_binCRTT_muon2016M_bin6
307 prop_binSR_muon2016M_bin17_TVX
308 prop_binCROS_muon2018_bin14
309 prop_binSR_electron2016M_bin17_WpWpJJ_QCD
310 prop_binCRTT_electron2016M_bin4
311 prop_binCROS_electron2018_bin13_Fake
312 prop_binSR_electron2017_bin17_WpWpJJ_QCD
313 prop_binCROS_electron2017_bin9_Fake
314 prop_binCROS_muon2017_bin10
315 prop_binSR_electron2016M_bin20_WpWpJJ_QCD
316 prop_binCROS_muon2018_bin7
317 prop_binSR_muon2017_bin0
318 prop_binCROS_electron2017_bin10_WrongSign
319 prop_binSR_muon2016M_bin20_Triboson
320 prop_binCROS_muon2017_bin6
321 prop_binCROS_muon2016M_bin7
322 prop_binCRTT_electron2017_bin2
323 prop_binSR_muon2016M_bin14_WpWpJJ_QCD
324 jer_2017
325 prop_binCRTT_muon2016M_bin5
326 prop_binCROS_electron2018_bin14_WrongSign
327 prop_binCRTT_muon2018_bin5
328 prop_binCROS_electron2018_bin16
329 prop_binCROS_electron2018_bin6_WrongSign
330 prop_binCRTT_electron2018_bin3
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

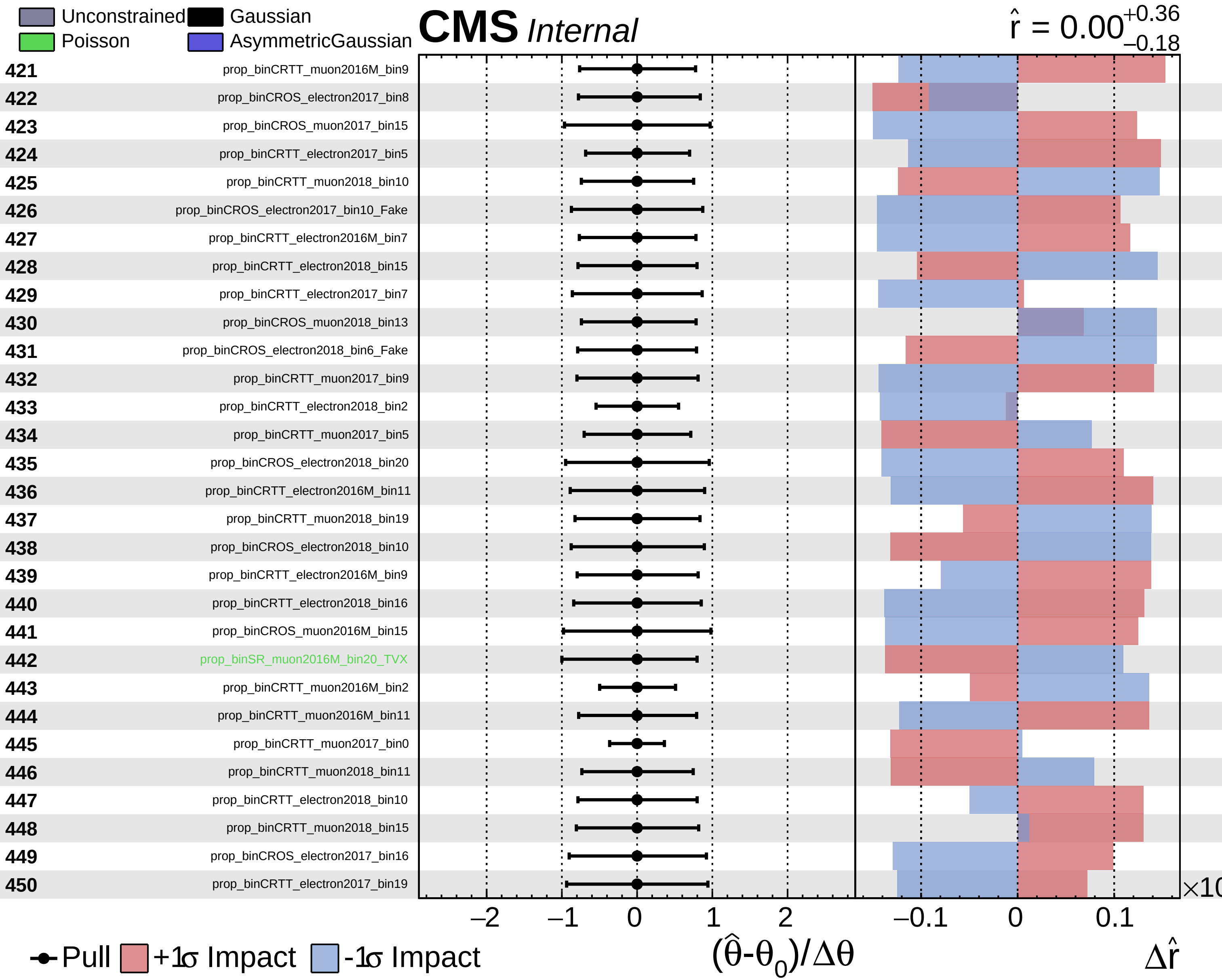


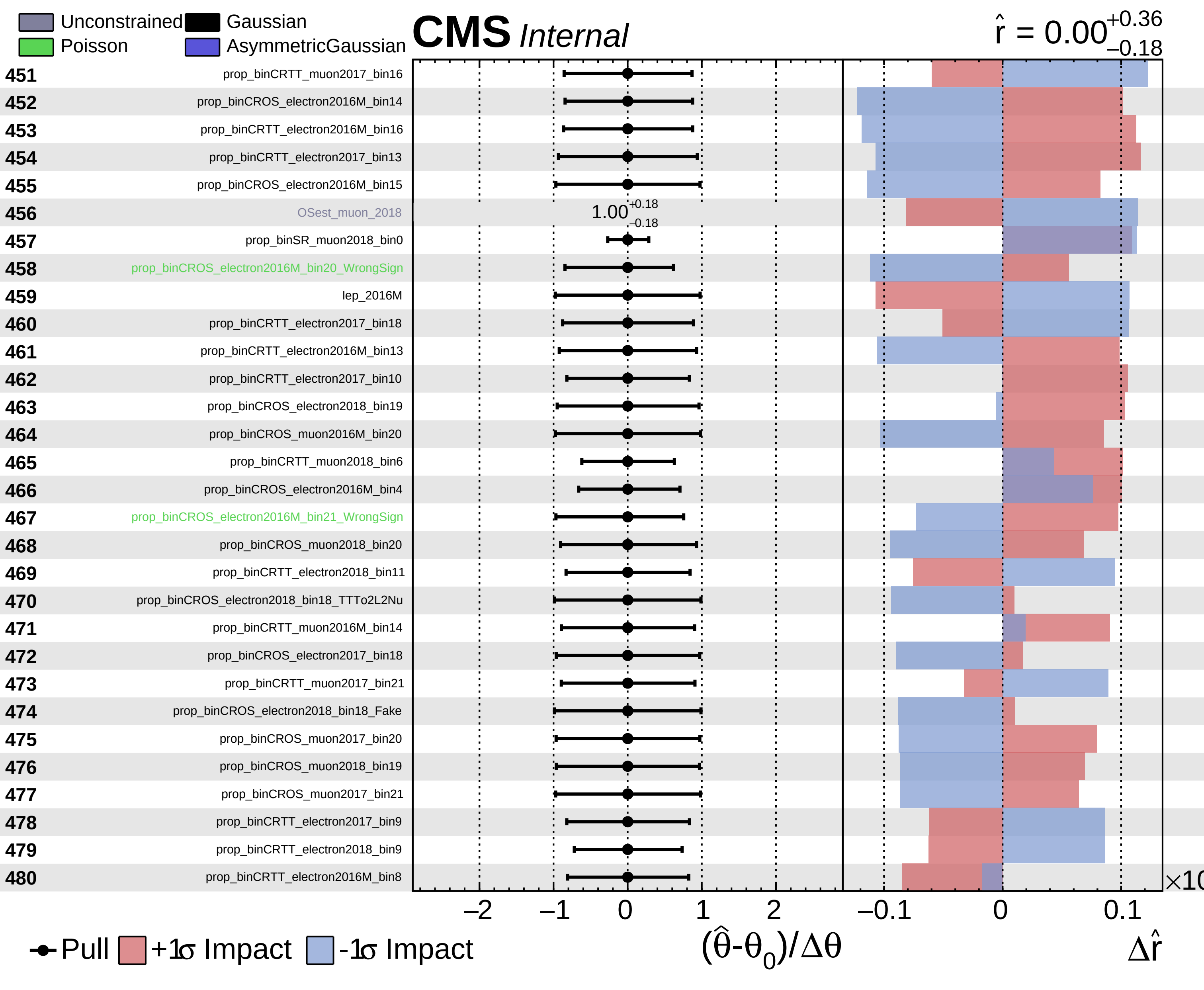
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.18}$
331 prop_binSR_electron2016M_bin20_TVX
332 prop_binCROS_muon2016M_bin8_Fake
333 QCDScale_TTdilep
334 prop_binSR_electron2016M_bin17_WrongSign
335 prop_binCRTT_electron2018_bin19
336 prop_binCRTT_electron2016M_bin1
337 prop_binCROS_electron2017_bin9_WrongSign
338 OSest_electron_2017
339 prop_binCRTT_muon2017_bin6
340 prop_binCROS_electron2018_bin21
341 prop_binSR_electron2017_bin20_WpWpJJ_QCD
342 prop_binCROS_muon2018_bin15
343 prop_binSR_electron2018_bin0
344 prop_binCRTT_electron2017_bin14
345 prop_binCROS_muon2017_bin5_WrongSign
346 prop_binCROS_electron2017_bin17
347 prop_binCROS_muon2018_bin17_WrongSign
348 prop_binCROS_muon2018_bin9_Fake
349 QCDScale_WrongSign
350 prop_binCRTT_muon2018_bin13
351 prop_binCRTT_electron2017_bin6
352 prop_binSR_electron2017_bin0
353 prop_binCRF_electron2016M_bin0
354 prop_binCROS_electron2018_bin9_Fake
355 prop_binCROS_muon2018_bin9_WrongSign
356 prop_binCROS_electron2018_bin15
357 prop_binSR_electron2016M_bin17_Triboson
358 prop_binCRTT_muon2017_bin14
359 prop_binCROS_muon2017_bin5_Fake
360 prop_binCRF_electron2017_bin0
$1.00^{+0.27}_{-0.27}$
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

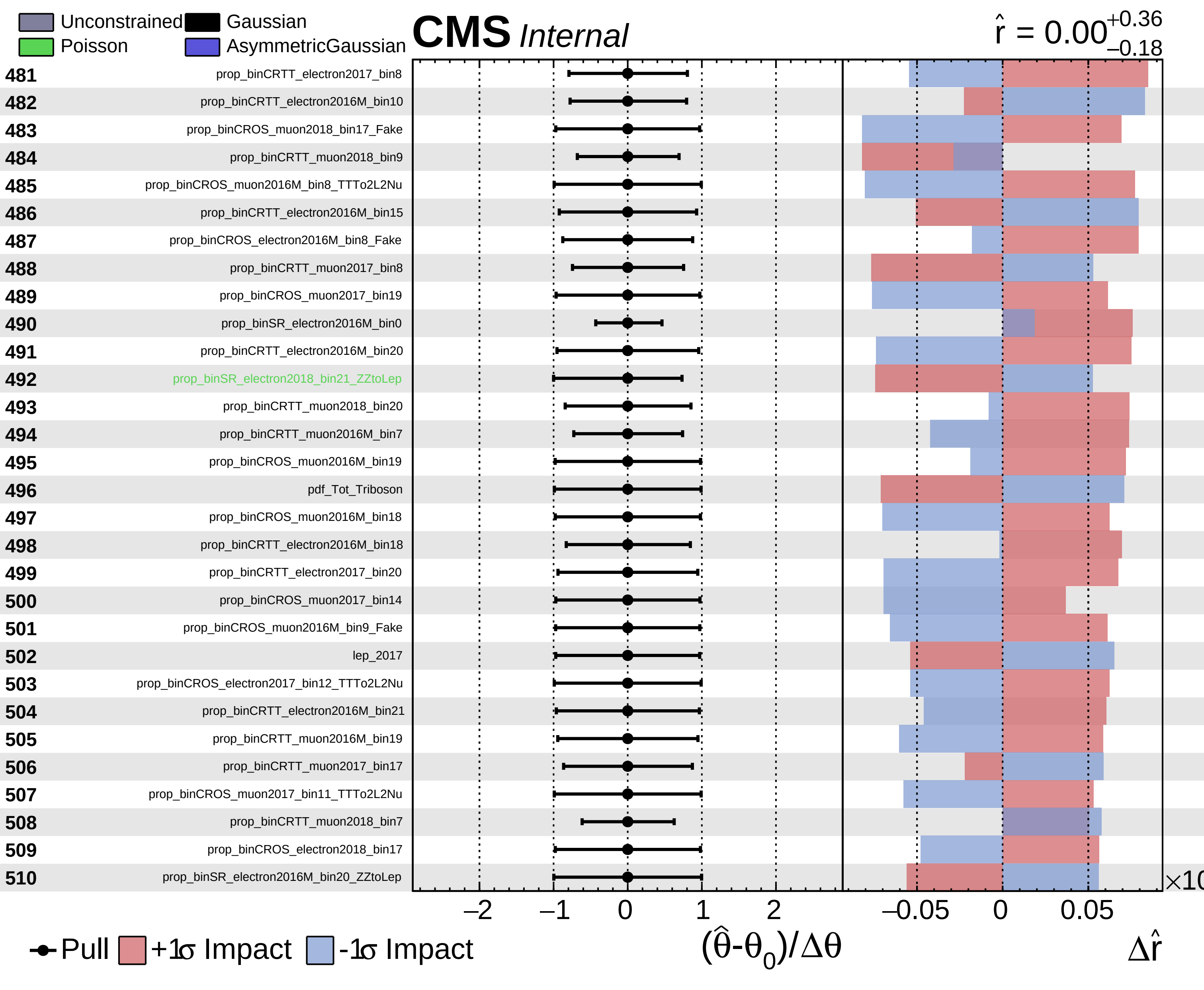

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.18}$
361 prop_binSR_electron2017_bin10_WpWpJJ_QCD
362 prop_binCRTT_electron2016M_bin6
363 prop_binCROS_electron2016M_bin10
364 prop_binCRTT_muon2018_bin14
365 prop_binCRTT_muon2018_bin4
366 prop_binSR_electron2016M_bin21_WpWpJJ_QCD
367 prop_binCRTT_electron2018_bin13
368 prop_binCRTT_muon2017_bin7
369 prop_binSR_electron2016M_bin21_Triboson
370 prop_binCRTT_muon2018_bin17
371 prop_binCROS_electron2018_bin9_WrongSign
372 prop_binCROS_muon2018_bin16
373 FSR_TTdilep
374 prop_binCRTT_muon2016M_bin16
375 prop_binCRTT_electron2017_bin17
376 prop_binCROS_muon2017_bin7
377 prop_binCRTT_electron2018_bin18
378 prop_binCRTT_muon2016M_bin17
379 prop_binCRTT_electron2017_bin16
380 prop_binCROS_electron2017_bin12_Fake
381 prop_binCRTT_electron2017_bin12
382 prop_binSR_electron2018_bin1_WrongSign
383 prop_binCRTT_muon2017_bin20
384 prop_binCRTT_electron2018_bin21
385 prop_binCRTT_muon2017_bin4
386 prop_binCRTT_muon2018_bin21
387 prop_binCRTT_electron2017_bin15
388 prop_binCRTT_electron2018_bin8
389 prop_binCRTT_muon2017_bin11
390 prop_binCROS_electron2017_bin15
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



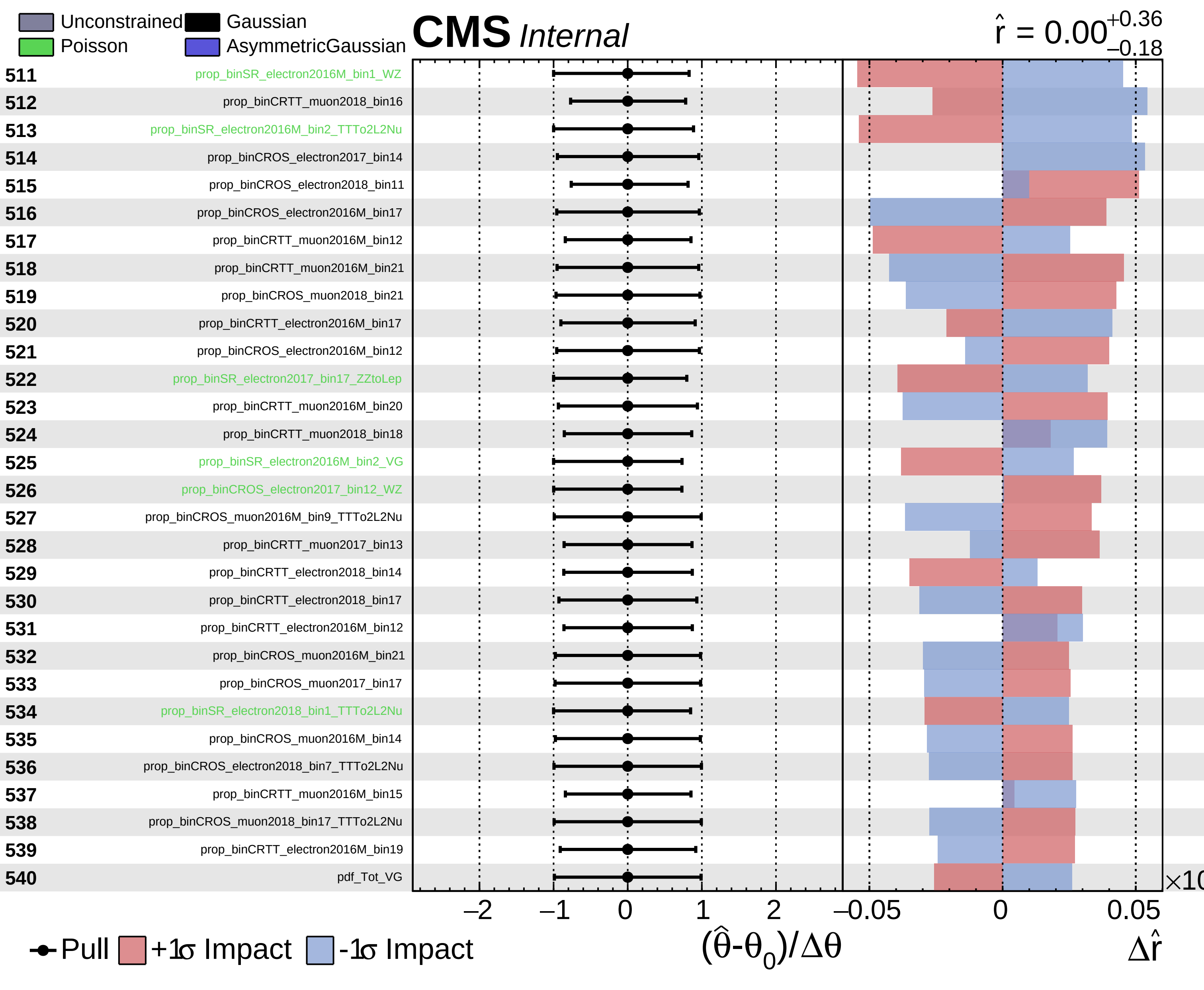
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.18}$
391 prop_binSR_muon2016M_bin14_TVX
392 prop_binCROS_electron2016M_bin7
393 prop_binCROS_muon2018_bin18
394 prop_binCRTT_muon2017_bin15
395 prop_binCRTT_muon2016M_bin4
396 prop_binSR_muon2016M_bin14_Triboson
397 prop_binCROS_electron2016M_bin6
398 prop_binCRTT_muon2018_bin12
399 prop_binCRTT_muon2017_bin18
400 prop_binSR_electron2017_bin17_Triboson
401 prop_binCROS_muon2018_bin11_Fake
402 prop_binCROS_electron2016M_bin8_WrongSign
403 prop_binCRTT_muon2016M_bin13
404 prop_binCRTT_electron2016M_bin14
405 prop_binCRTT_muon2017_bin2
406 prop_binCRTT_electron2017_bin3
407 prop_binCRTT_electron2016M_bin3
408 prop_binSR_electron2017_bin10_TVX
409 prop_binCRTT_muon2016M_bin18
410 prop_binCRTT_electron2016M_bin5
411 prop_binCROS_muon2016M_bin17
412 prop_binCRTT_muon2018_bin8
413 prop_binCRTT_muon2017_bin1
414 prop_binCROS_electron2017_bin13
415 prop_binCROS_electron2016M_bin13
416 lep_2018
417 prop_binCROS_electron2016M_bin9_WrongSign
418 prop_binCROS_muon2016M_bin12
419 prop_binCROS_muon2016M_bin13
420 prop_binCROS_muon2017_bin11_Fake
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

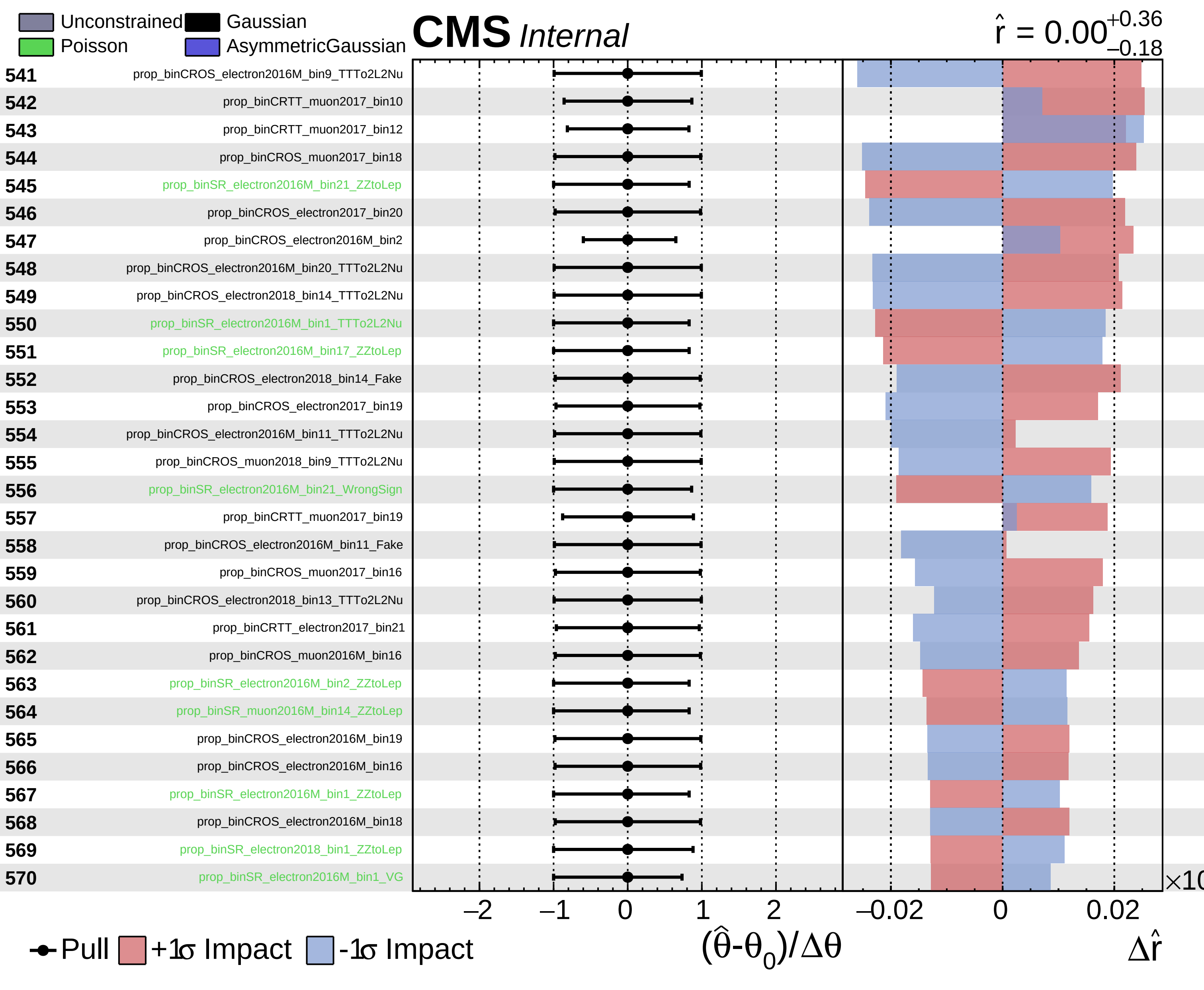

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.18}$
421 prop_binCRTT_muon2016M_bin9
422 prop_binCROS_electron2017_bin8
423 prop_binCROS_muon2017_bin15
424 prop_binCRTT_electron2017_bin5
425 prop_binCRTT_muon2018_bin10
426 prop_binCROS_electron2017_bin10_Fake
427 prop_binCRTT_electron2016M_bin7
428 prop_binCRTT_electron2018_bin15
429 prop_binCRTT_electron2017_bin7
430 prop_binCROS_muon2018_bin13
431 prop_binCROS_electron2018_bin6_Fake
432 prop_binCRTT_muon2017_bin9
433 prop_binCRTT_electron2018_bin2
434 prop_binCRTT_muon2017_bin5
435 prop_binCROS_electron2018_bin20
436 prop_binCRTT_electron2016M_bin11
437 prop_binCRTT_muon2018_bin19
438 prop_binCROS_electron2018_bin10
439 prop_binCRTT_electron2016M_bin9
440 prop_binCRTT_electron2018_bin16
441 prop_binCROS_muon2016M_bin15
442 prop_binSR_muon2016M_bin20_TVX
443 prop_binCRTT_muon2016M_bin2
444 prop_binCRTT_muon2016M_bin11
445 prop_binCRTT_muon2017_bin0
446 prop_binCRTT_muon2018_bin11
447 prop_binCRTT_electron2018_bin10
448 prop_binCRTT_muon2018_bin15
449 prop_binCROS_electron2017_bin16
450 prop_binCRTT_electron2017_bin19
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

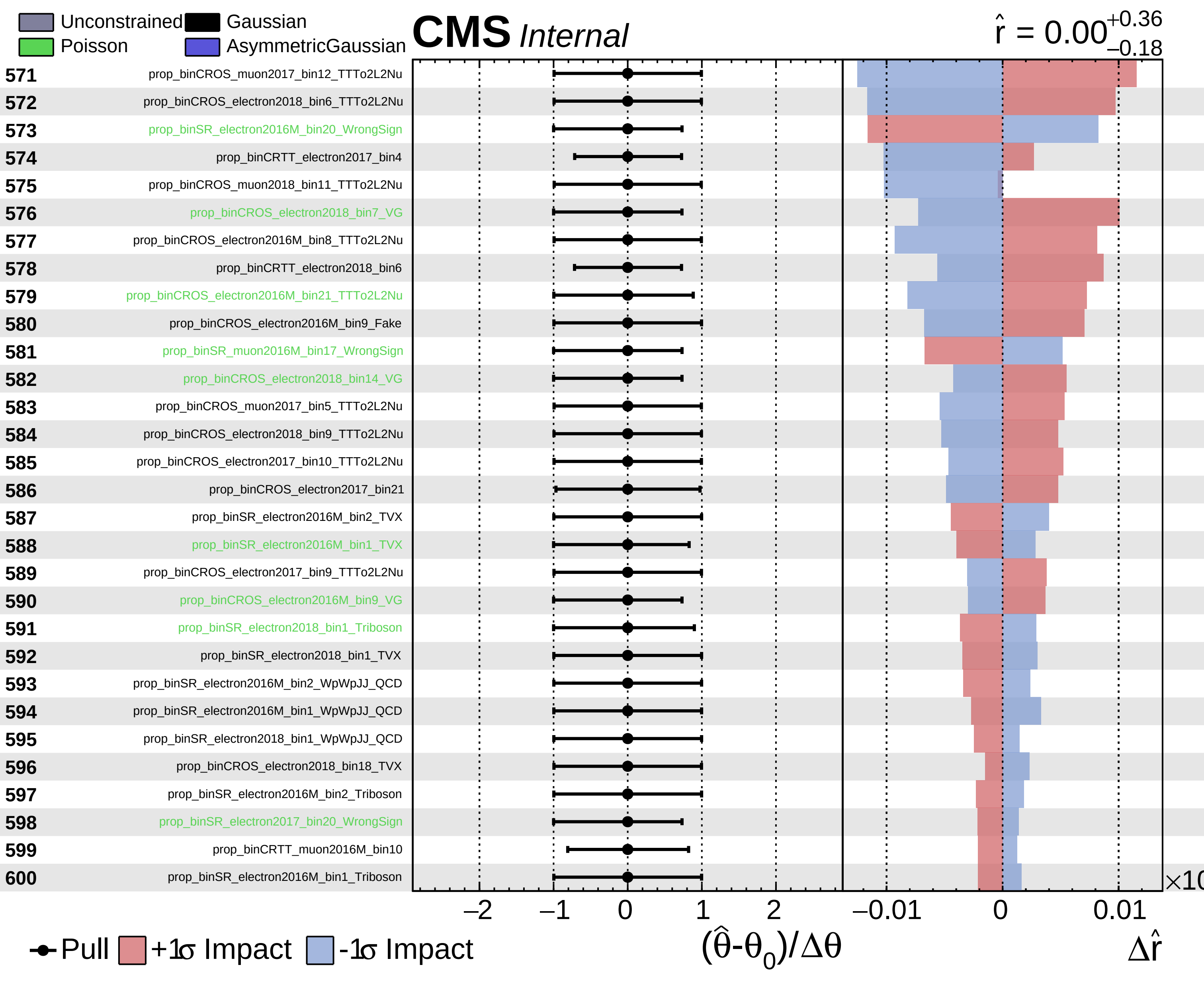

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.18}$
451 prop_binCRTT_muon2017_bin16
452 prop_binCROS_electron2016M_bin14
453 prop_binCRTT_electron2016M_bin16
454 prop_binCRTT_electron2017_bin13
455 prop_binCROS_electron2016M_bin15
456 OSest_muon_2018
$1.00^{+0.18}_{-0.18}$
457 prop_binSR_muon2018_bin0
458 prop_binCROS_electron2016M_bin20_WrongSign
459 lep_2016M
460 prop_binCRTT_electron2017_bin18
461 prop_binCRTT_electron2016M_bin13
462 prop_binCRTT_electron2017_bin10
463 prop_binCROS_electron2018_bin19
464 prop_binCROS_muon2016M_bin20
465 prop_binCRTT_muon2018_bin6
466 prop_binCROS_electron2016M_bin4
467 prop_binCROS_electron2016M_bin21_WrongSign
468 prop_binCROS_muon2018_bin20
469 prop_binCRTT_electron2018_bin11
470 prop_binCROS_electron2018_bin18_TTTo2L2Nu
471 prop_binCRTT_muon2016M_bin14
472 prop_binCROS_electron2017_bin18
473 prop_binCRTT_muon2017_bin21
474 prop_binCROS_electron2018_bin18_Fake
475 prop_binCROS_muon2017_bin20
476 prop_binCROS_muon2018_bin19
477 prop_binCROS_muon2017_bin21
478 prop_binCRTT_electron2017_bin9
479 prop_binCRTT_electron2018_bin9
480 prop_binCRTT_electron2016M_bin8
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



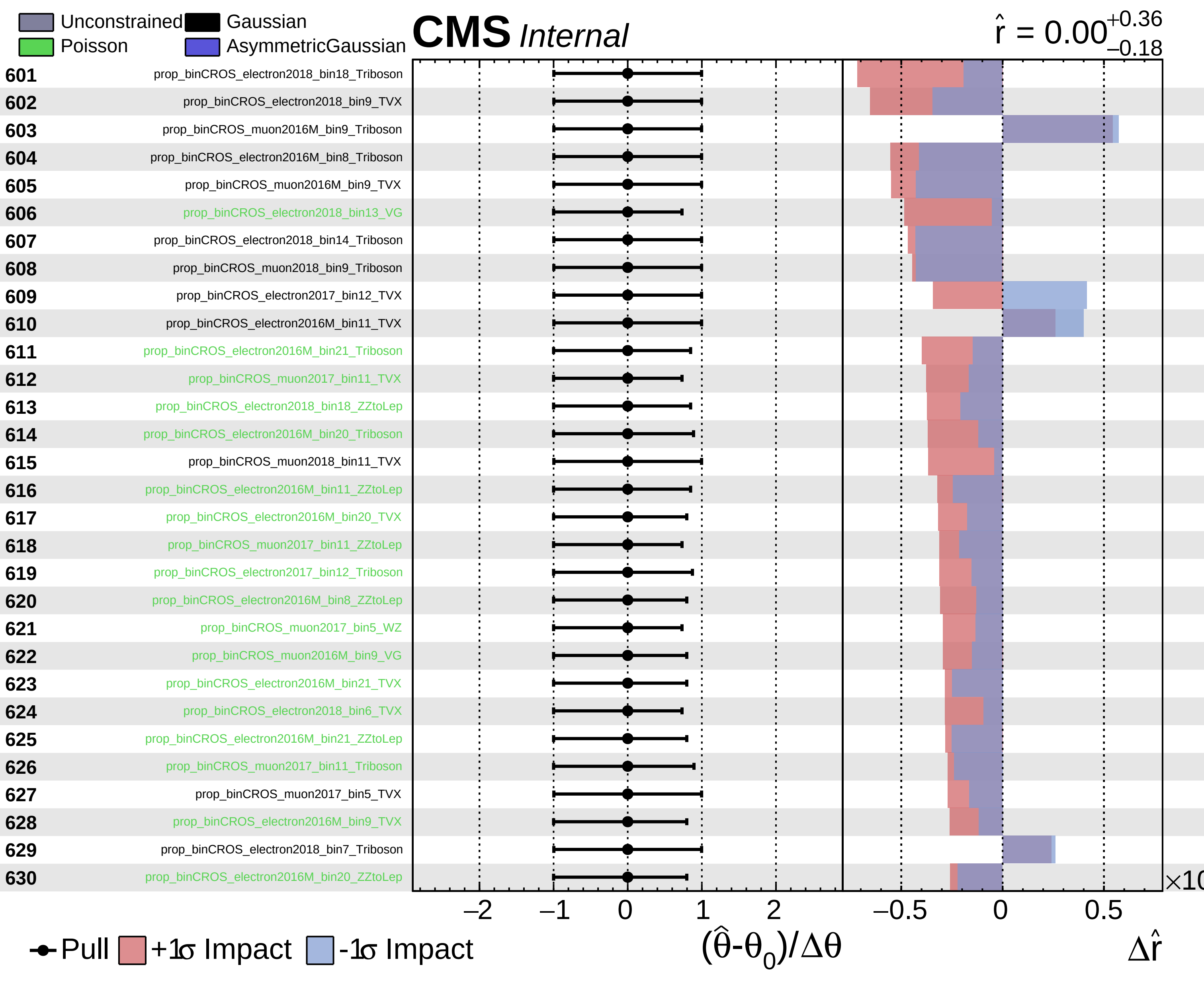
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.18}$
481 prop_binCRTT_electron2017_bin8
482 prop_binCRTT_electron2016M_bin10
483 prop_binCROS_muon2018_bin17_Fake
484 prop_binCRTT_muon2018_bin9
485 prop_binCROS_muon2016M_bin8_TTTo2L2Nu
486 prop_binCRTT_electron2016M_bin15
487 prop_binCROS_electron2016M_bin8_Fake
488 prop_binCRTT_muon2017_bin8
489 prop_binCROS_muon2017_bin19
490 prop_binSR_electron2016M_bin0
491 prop_binCRTT_electron2016M_bin20
492 prop_binSR_electron2018_bin21_ZZtoLep
493 prop_binCRTT_muon2018_bin20
494 prop_binCRTT_muon2016M_bin7
495 prop_binCROS_muon2016M_bin19
496 pdf_Tot_Triboson
497 prop_binCROS_muon2016M_bin18
498 prop_binCRTT_electron2016M_bin18
499 prop_binCRTT_electron2017_bin20
500 prop_binCROS_muon2017_bin14
501 prop_binCROS_muon2016M_bin9_Fake
502 lep_2017
503 prop_binCROS_electron2017_bin12_TTTo2L2Nu
504 prop_binCRTT_electron2016M_bin21
505 prop_binCRTT_muon2016M_bin19
506 prop_binCRTT_muon2017_bin17
507 prop_binCROS_muon2017_bin11_TTTo2L2Nu
508 prop_binCRTT_muon2018_bin7
509 prop_binCROS_electron2018_bin17
510 prop_binSR_electron2016M_bin20_ZZtoLep
−2
−1
0
1
2
−0.05
0
0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.18}$
511 prop_binSR_electron2016M_bin1_WZ
512 prop_binCRTT_muon2018_bin16
513 prop_binSR_electron2016M_bin2_TTTo2L2Nu
514 prop_binCROS_electron2017_bin14
515 prop_binCROS_electron2018_bin11
516 prop_binCROS_electron2016M_bin17
517 prop_binCRTT_muon2016M_bin12
518 prop_binCRTT_muon2016M_bin21
519 prop_binCROS_muon2018_bin21
520 prop_binCRTT_electron2016M_bin17
521 prop_binCROS_electron2016M_bin12
522 prop_binSR_electron2017_bin17_ZZtoLep
523 prop_binCRTT_muon2016M_bin20
524 prop_binCRTT_muon2018_bin18
525 prop_binSR_electron2016M_bin2_VG
526 prop_binCROS_electron2017_bin12_WZ
527 prop_binCROS_muon2016M_bin9_TTTo2L2Nu
528 prop_binCRTT_muon2017_bin13
529 prop_binCRTT_electron2018_bin14
530 prop_binCRTT_electron2018_bin17
531 prop_binCRTT_electron2016M_bin12
532 prop_binCROS_muon2016M_bin21
533 prop_binCROS_muon2017_bin17
534 prop_binSR_electron2018_bin1_TTTo2L2Nu
535 prop_binCROS_muon2016M_bin14
536 prop_binCROS_electron2018_bin7_TTTo2L2Nu
537 prop_binCRTT_muon2016M_bin15
538 prop_binCROS_muon2018_bin17_TTTo2L2Nu
539 prop_binCRTT_electron2016M_bin19
540 pdf_Tot_VG
−2
−1
0
1
2
−0.05
0
0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

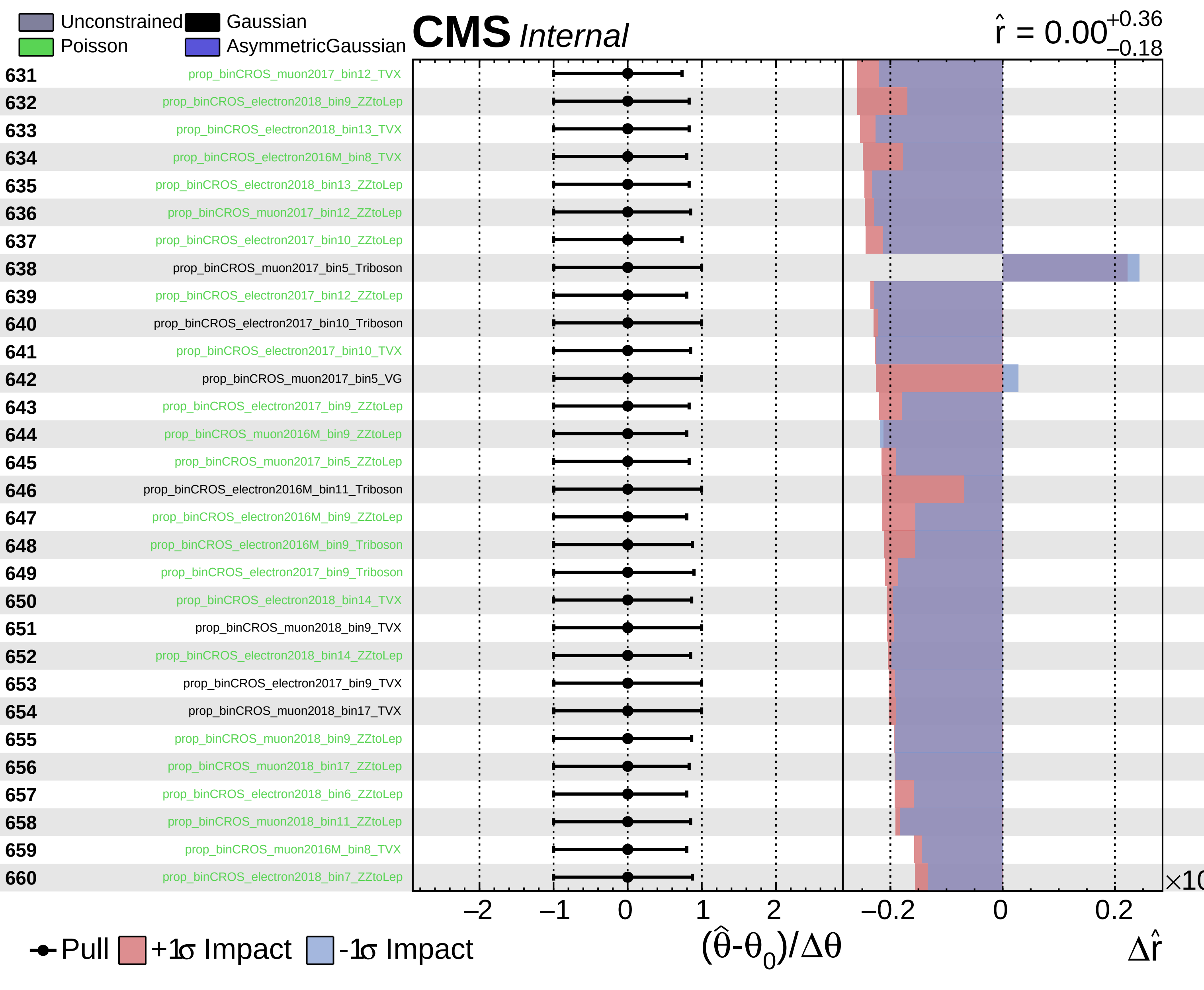

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.18}$
541 prop_binCROS_electron2016M_bin9_TTTo2L2Nu
542 prop_binCRTT_muon2017_bin10
543 prop_binCRTT_muon2017_bin12
544 prop_binCROS_muon2017_bin18
545 prop_binSR_electron2016M_bin21_ZZtoLep
546 prop_binCROS_electron2017_bin20
547 prop_binCROS_electron2016M_bin2
548 prop_binCROS_electron2016M_bin20_TTTo2L2Nu
549 prop_binCROS_electron2018_bin14_TTTo2L2Nu
550 prop_binSR_electron2016M_bin1_TTTo2L2Nu
551 prop_binSR_electron2016M_bin17_ZZtoLep
552 prop_binCROS_electron2018_bin14_Fake
553 prop_binCROS_electron2017_bin19
554 prop_binCROS_electron2016M_bin11_TTTo2L2Nu
555 prop_binCROS_muon2018_bin9_TTTo2L2Nu
556 prop_binSR_electron2016M_bin21_WrongSign
557 prop_binCRTT_muon2017_bin19
558 prop_binCROS_electron2016M_bin11_Fake
559 prop_binCROS_muon2017_bin16
560 prop_binCROS_electron2018_bin13_TTTo2L2Nu
561 prop_binCRTT_electron2017_bin21
562 prop_binCROS_muon2016M_bin16
563 prop_binSR_electron2016M_bin2_ZZtoLep
564 prop_binSR_muon2016M_bin14_ZZtoLep
565 prop_binCROS_electron2016M_bin19
566 prop_binCROS_electron2016M_bin16
567 prop_binSR_electron2016M_bin1_ZZtoLep
568 prop_binCROS_electron2016M_bin18
569 prop_binSR_electron2018_bin1_ZZtoLep
570 prop_binSR_electron2016M_bin1_VG
−2
−1
0
1
2
−0.02
0
0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



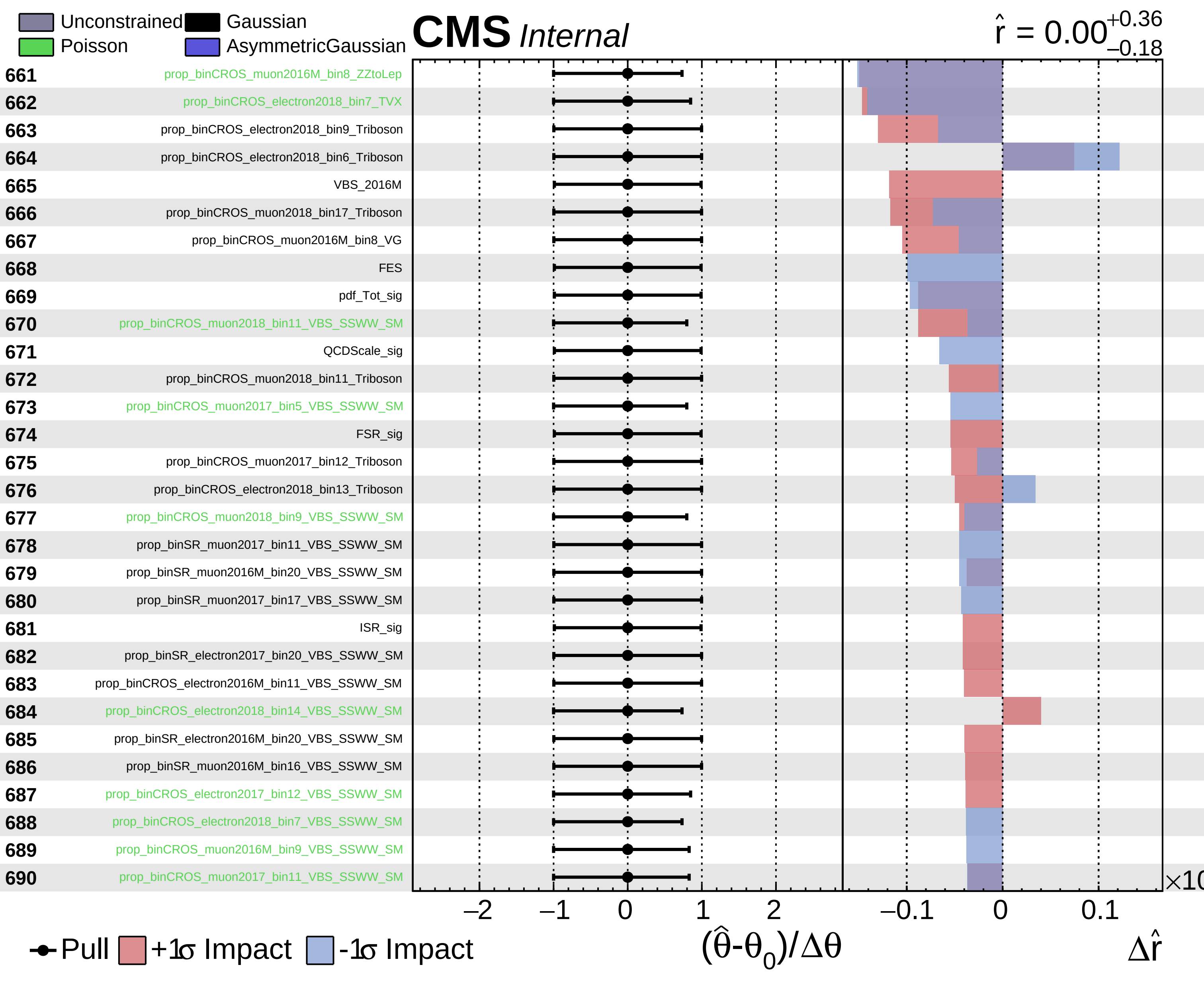
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.36 −0.18
571 prop_binCROS_muon2017_bin12_TTTo2L2Nu
572 prop_binCROS_electron2018_bin6_TTTo2L2Nu
573 prop_binSR_electron2016M_bin20_WrongSign
574 prop_binCRTT_electron2017_bin4
575 prop_binCROS_muon2018_bin11_TTTo2L2Nu
576 prop_binCROS_electron2018_bin7_VG
577 prop_binCROS_electron2016M_bin8_TTTo2L2Nu
578 prop_binCRTT_electron2018_bin6
579 prop_binCROS_electron2016M_bin21_TTTo2L2Nu
580 prop_binCROS_electron2016M_bin9_Fake
581 prop_binSR_muon2016M_bin17_WrongSign
582 prop_binCROS_electron2018_bin14_VG
583 prop_binCROS_muon2017_bin5_TTTo2L2Nu
584 prop_binCROS_electron2018_bin9_TTTo2L2Nu
585 prop_binCROS_electron2017_bin10_TTTo2L2Nu
586 prop_binCROS_electron2017_bin21
587 prop_binSR_electron2016M_bin2_TVX
588 prop_binSR_electron2016M_bin1_TVX
589 prop_binCROS_electron2017_bin9_TTTo2L2Nu
590 prop_binCROS_electron2016M_bin9_VG
591 prop_binSR_electron2018_bin1_Triboson
592 prop_binSR_electron2018_bin1_TVX
593 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
594 prop_binSR_electron2016M_bin1_WpWpJJ_QCD
595 prop_binSR_electron2018_bin1_WpWpJJ_QCD
596 prop_binCROS_electron2018_bin18_TVX
597 prop_binSR_electron2016M_bin2_Triboson
598 prop_binSR_electron2017_bin20_WrongSign
599 prop_binCRTT_muon2016M_bin10
600 prop_binSR_electron2016M_bin1_Triboson
−2
−1
0
1
2
−0.01
0
0.01
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



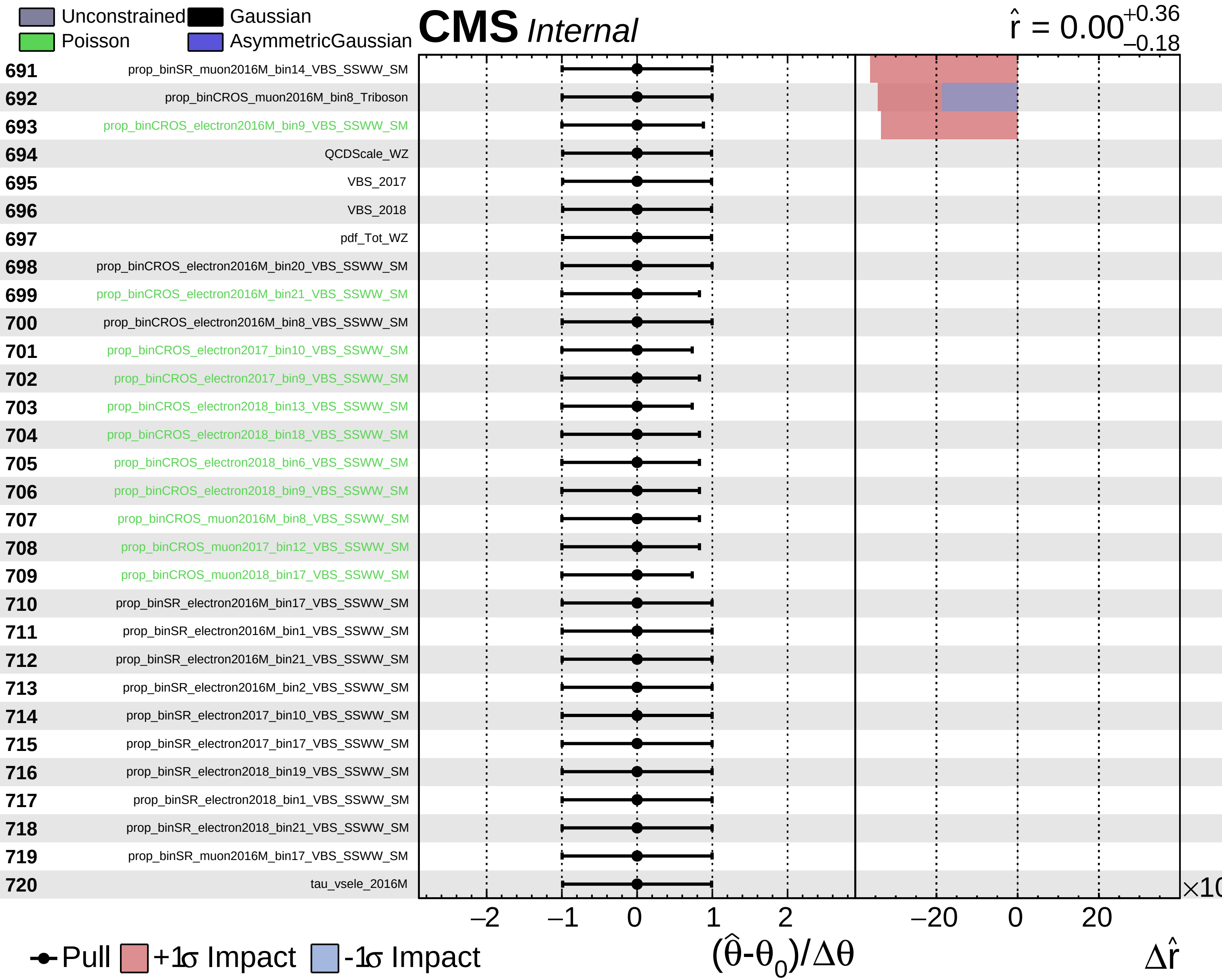
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.18}$
601 prop_binCROS_electron2018_bin18_Triboson
602 prop_binCROS_electron2018_bin9_TVX
603 prop_binCROS_muon2016M_bin9_Triboson
604 prop_binCROS_electron2016M_bin8_Triboson
605 prop_binCROS_muon2016M_bin9_TVX
606 prop_binCROS_electron2018_bin13_VG
607 prop_binCROS_electron2018_bin14_Triboson
608 prop_binCROS_muon2018_bin9_Triboson
609 prop_binCROS_electron2017_bin12_TVX
610 prop_binCROS_electron2016M_bin11_TVX
611 prop_binCROS_electron2016M_bin21_Triboson
612 prop_binCROS_muon2017_bin11_TVX
613 prop_binCROS_electron2018_bin18_ZZtoLep
614 prop_binCROS_electron2016M_bin20_Triboson
615 prop_binCROS_muon2018_bin11_TVX
616 prop_binCROS_electron2016M_bin11_ZZtoLep
617 prop_binCROS_electron2016M_bin20_TVX
618 prop_binCROS_muon2017_bin11_ZZtoLep
619 prop_binCROS_electron2017_bin12_Triboson
620 prop_binCROS_electron2016M_bin8_ZZtoLep
621 prop_binCROS_muon2017_bin5_WZ
622 prop_binCROS_muon2016M_bin9_VG
623 prop_binCROS_electron2016M_bin21_TVX
624 prop_binCROS_electron2018_bin6_TVX
625 prop_binCROS_electron2016M_bin21_ZZtoLep
626 prop_binCROS_muon2017_bin11_Triboson
627 prop_binCROS_muon2017_bin5_TVX
628 prop_binCROS_electron2016M_bin9_TVX
629 prop_binCROS_electron2018_bin7_Triboson
630 prop_binCROS_electron2016M_bin20_ZZtoLep
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



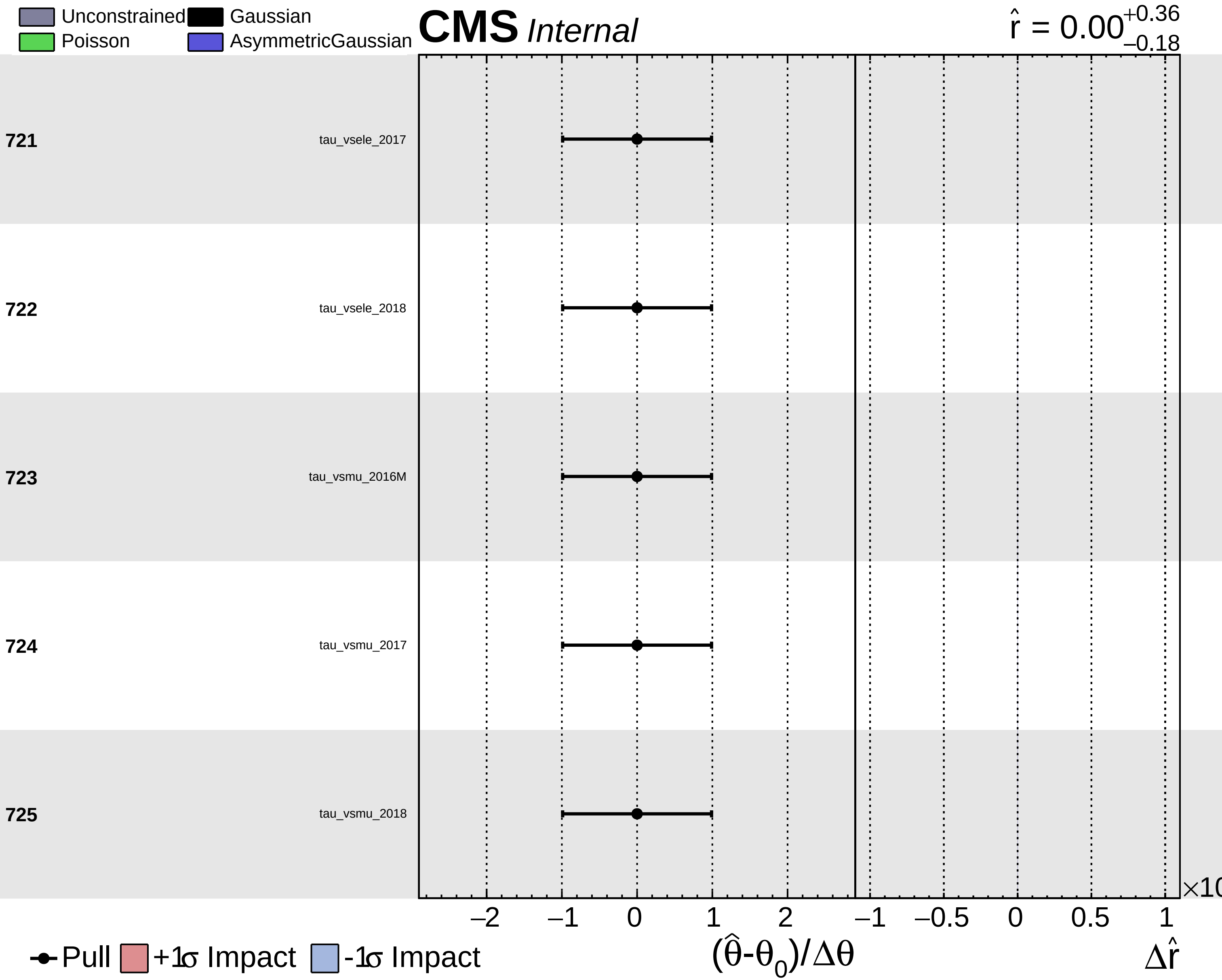
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.18}$
631 prop_binCROS_muon2017_bin12_TVX
632 prop_binCROS_electron2018_bin9_ZZtoLep
633 prop_binCROS_electron2018_bin13_TVX
634 prop_binCROS_electron2016M_bin8_TVX
635 prop_binCROS_electron2018_bin13_ZZtoLep
636 prop_binCROS_muon2017_bin12_ZZtoLep
637 prop_binCROS_electron2017_bin10_ZZtoLep
638 prop_binCROS_muon2017_bin5_Triboson
639 prop_binCROS_electron2017_bin12_ZZtoLep
640 prop_binCROS_electron2017_bin10_Triboson
641 prop_binCROS_electron2017_bin10_TVX
642 prop_binCROS_muon2017_bin5_VG
643 prop_binCROS_electron2017_bin9_ZZtoLep
644 prop_binCROS_muon2016M_bin9_ZZtoLep
645 prop_binCROS_muon2017_bin5_ZZtoLep
646 prop_binCROS_electron2016M_bin11_Triboson
647 prop_binCROS_electron2016M_bin9_ZZtoLep
648 prop_binCROS_electron2016M_bin9_Triboson
649 prop_binCROS_electron2017_bin9_Triboson
650 prop_binCROS_electron2018_bin14_TVX
651 prop_binCROS_muon2018_bin9_TVX
652 prop_binCROS_electron2018_bin14_ZZtoLep
653 prop_binCROS_electron2017_bin9_TVX
654 prop_binCROS_muon2018_bin17_TVX
655 prop_binCROS_muon2018_bin9_ZZtoLep
656 prop_binCROS_muon2018_bin17_ZZtoLep
657 prop_binCROS_electron2018_bin6_ZZtoLep
658 prop_binCROS_muon2018_bin11_ZZtoLep
659 prop_binCROS_muon2016M_bin8_TVX
660 prop_binCROS_electron2018_bin7_ZZtoLep
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.18}$
661 prop_binCROS_muon2016M_bin8_ZZtoLep
662 prop_binCROS_electron2018_bin7_TVX
663 prop_binCROS_electron2018_bin9_Triboson
664 prop_binCROS_electron2018_bin6_Triboson
665 VBS_2016M
666 prop_binCROS_muon2018_bin17_Triboson
667 prop_binCROS_muon2016M_bin8_VG
668 FES
669 pdf_Tot_sig
670 prop_binCROS_muon2018_bin11_VBS_SSWW_SM
671 QCDScale_sig
672 prop_binCROS_muon2018_bin11_Triboson
673 prop_binCROS_muon2017_bin5_VBS_SSWW_SM
674 FSR_sig
675 prop_binCROS_muon2017_bin12_Triboson
676 prop_binCROS_electron2018_bin13_Triboson
677 prop_binCROS_muon2018_bin9_VBS_SSWW_SM
678 prop_binSR_muon2017_bin11_VBS_SSWW_SM
679 prop_binSR_muon2016M_bin20_VBS_SSWW_SM
680 prop_binSR_muon2017_bin17_VBS_SSWW_SM
681 ISR_sig
682 prop_binSR_electron2017_bin20_VBS_SSWW_SM
683 prop_binCROS_electron2016M_bin11_VBS_SSWW_SM
684 prop_binCROS_electron2018_bin14_VBS_SSWW_SM
685 prop_binSR_electron2016M_bin20_VBS_SSWW_SM
686 prop_binSR_muon2016M_bin16_VBS_SSWW_SM
687 prop_binCROS_electron2017_bin12_VBS_SSWW_SM
688 prop_binCROS_electron2018_bin7_VBS_SSWW_SM
689 prop_binCROS_muon2016M_bin9_VBS_SSWW_SM
690 prop_binCROS_muon2017_bin11_VBS_SSWW_SM
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.36 −0.18
691 prop_binSR_muon2016M_bin14_VBS_SSWW_SM
692 prop_binCROS_muon2016M_bin8_Triboson
693 prop_binCROS_electron2016M_bin9_VBS_SSWW_SM
694 QCDScale_WZ
695 VBS_2017
696 VBS_2018
697 pdf_Tot_WZ
698 prop_binCROS_electron2016M_bin20_VBS_SSWW_SM
699 prop_binCROS_electron2016M_bin21_VBS_SSWW_SM
700 prop_binCROS_electron2016M_bin8_VBS_SSWW_SM
701 prop_binCROS_electron2017_bin10_VBS_SSWW_SM
702 prop_binCROS_electron2017_bin9_VBS_SSWW_SM
703 prop_binCROS_electron2018_bin13_VBS_SSWW_SM
704 prop_binCROS_electron2018_bin18_VBS_SSWW_SM
705 prop_binCROS_electron2018_bin6_VBS_SSWW_SM
706 prop_binCROS_electron2018_bin9_VBS_SSWW_SM
707 prop_binCROS_muon2016M_bin8_VBS_SSWW_SM
708 prop_binCROS_muon2017_bin12_VBS_SSWW_SM
709 prop_binCROS_muon2018_bin17_VBS_SSWW_SM
710 prop_binSR_electron2016M_bin17_VBS_SSWW_SM
711 prop_binSR_electron2016M_bin1_VBS_SSWW_SM
712 prop_binSR_electron2016M_bin21_VBS_SSWW_SM
713 prop_binSR_electron2016M_bin2_VBS_SSWW_SM
714 prop_binSR_electron2017_bin10_VBS_SSWW_SM
715 prop_binSR_electron2017_bin17_VBS_SSWW_SM
716 prop_binSR_electron2018_bin19_VBS_SSWW_SM
717 prop_binSR_electron2018_bin1_VBS_SSWW_SM
718 prop_binSR_electron2018_bin21_VBS_SSWW_SM
719 prop_binSR_muon2016M_bin17_VBS_SSWW_SM
720 tau_vsele_2016M
−2
−1
0
1
2
−20
0
20
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.18}$
721
tau_vsele_2017
722
tau_vsele_2018
723
tau_vsmu_2016M
724
tau_vsmu_2017
725
tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$